SecondChance Supplement: Exact Structural Schedule Frontiers and Preserved Identification Proofs

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v21 exact saved-stream schedule analysis plus unchanged v20 sections and inherited full v19 manuscript. No independent natural evidence or actual audit added.

Abstract

Detailed assumptions, derivations, full comparison contracts, negative evidence and reproducibility notes accompany the compact main paper. The full previous scientific record is retained without changing its reported numbers or evidence scope. New finite computations reuse existing exposed evidence; no model invocation or independent empirical sample is introduced. New v21 sections distinguish these calculations from preserved v20 explanations and the v19 scientific record.

1 S1. Reading route and the recovery contract

The compact manuscript follows an identification question rather than a software-development chronology. Read the endpoint definitions before interpreting a recovery count, the information contract before interpreting a policy, and the denominator before interpreting an area. Sections S2-S7 establish the mathematical contract; S8-S11 connect it to the frozen replay; S12-S13 delimit verification and the next admissible experiment. S14 routes the preserved full manuscript without presenting historical evidence as new confirmation.

The requirement remains fixed when an evaluator is corrected. A requirement change, an unapproved correction or an unavailable archived answer is a different eligibility event. E1 rescoring reuses an old answer; E2 executes a candidate in its own context; E3 executes the integrated current artifact with the surviving lesson and preserved later changes. Agreement of E1 and E2 does not imply E3. Returning a historical verdict to the right value is neither an integration action nor an observation of the current state.

The historical B8 fixture has twelve rescoring improvements and no integrated recoveries from rescoring alone. Selection procedures find twelve development-positive candidates but four initial-state successes. At B5 the strong neutral joint control ties selective recovery with lower work and collateral loss. These retained adverse controls motivate auditing information rather than promoting a repair policy. None is a natural incidence estimate. The historical wording, tables and source-review limits remain unchanged in inherited.paper.

2 S2. Notation, signed occupation and timing

Fix one of the three utility panels and the same input whenever an identity is reused. A complete identity includes everything on which that panel utility depends. Two identical strings do not prove that the environment is complete or the executor deterministic. Conditional on these assumptions, one state utility is shared across arm visits, templates and checkpoints. This equality constraint can make direct paired bounds tighter than subtracting unrelated area intervals.

The neutral-minus-selective target averages twelve paired normalized utility integrals. Completed-prefix coefficient p integrates [0,t), retaining denominator 12B even when t is small. Current-frequency difference d is an instantaneous arm-occupation difference, not a completed integral. Both vectors sum to zero over a complete identity universe. Remaining fraction r=(B-t)/B is added on the original scale rather than renormalizing the prefix to one.

Commits through integer checkpoint t are revealed before a possible quote; no quote occurs at t=0. Recovery proceeds independently of auditing. A quote at t=B can narrow the final comparison but occupies no positive interval in the left-step width area. Query cap q is a separate upper bound. Dividing area by q, charging a cache quote as an actual worker call, or deleting the unknown suffix after inspecting every currently visible identity would change the experiment.

$$D = \frac{1}{12B} \sum_{h=1}^{12} \int_0^B \big(U\big(S_{h,N}(z)\big) - U\big(S_{h,L}(z)\big) \big)\, \mathrm{d}z \tag{1}$$

$$p_s(t) = \frac{1}{12B} \sum_{h=1}^{12} \int_0^t \big(\mathbf{1}\big[S_{h,N}(z) = s\big] - \mathbf{1}\big[S_{h,L}(z) = s\big] \big)\, \mathrm{d}z \tag{2}$$

$$r_t = \frac{B-t}{B}, \quad c(\lambda) = p + \lambda r_t d \tag{3}$$

Symbol	Meaning	Permission
B,t,r	Horizon, checkpoint, remaining fraction	Public
p,d	Prefix and current arm difference	Arrived commits only
A	Quoted identities / planned audit set	Identities enter selector
u_s	Fixed utility in [0,1]	Bound evaluator after quote
c0,c1	Restricted segment endpoints	Public construction
W_t,J	Safe width / normalized area	Evaluation

Table 1: A mathematical variable can exist in the evaluator while remaining unavailable to the selector.

3 S3. Exact scenario-conditioned and scenario-union intervals

For a fixed coefficient vector, known contributions are fixed and each unknown utility lies independently in [0,1]. A positive coefficient is minimized at zero and maximized at one; a negative coefficient makes the opposite choices. Coordinatewise extrema give Li and Ui below. Their difference is the unknown absolute-coefficient mass, and each endpoint is attainable in this audit-only box model. Additional semantic or joint-panel restrictions can shrink the box and make these endpoints conservative for that stronger model.

Along c(lambda), the unknown absolute-coefficient mass is convex, so its maximum on the closed segment occurs at lambda zero or one. This proves W_cond, the worst width after the scenario is fixed. It excludes movement of the known contribution between scenarios. Auditing every original state can therefore make W_cond zero while leaving the final coefficient vector unknown.

Now fix revealed utilities and vary both lambda and the unknown utilities. The lower endpoint is linear plus concave min terms in lambda; its minimum occurs at an endpoint. The upper endpoint is linear plus convex max terms; its maximum also occurs at an endpoint. The parameter domain is connected and the contrast is a continuous scalar map. Its attainable union is thus an interval, not merely two disconnected endpoint intervals, with extremes min(L0,L1) and max(U0,U1). W_union is exact under the segment and utility-box assumptions and is at least W_cond.

The order of operations matters: taking a within-scenario range and then its maximum differs from taking a range across all scenarios. Calling both robustness erases which uncertainty remains. The realized premium W_union-W_cond is nonnegative but need not be positive on a particular saved panel.

$$L_i = \sum_{s \in A} c_{i,s} u_s + \sum_{s \in A^c} \min\bigl(c_{i,s}, 0\bigr) \tag{4}$$

$$U_i = \sum_{s \in A} c_{i,s} u_s + \sum_{s \in A^c} \max\bigl(c_{i,s}, 0\bigr) \tag{5}$$

$$W_{\mathrm{cond}}(A) = \max_{i \in \{0,1\}} \sum_{s \in A^c} \bigl|c_{i,s}\bigr| \tag{6}$$

$$W_{\mathrm{union}}(A; u_A) = \max_{i \in \{0,1\}} U_i - \min_{i \in \{0,1\}} L_i \tag{7}$$

4 S4. Deriving the four-pair pre-outcome objective

The selector is deliberately utility-blind, even to values from previous quotes. Queried identities are retained, while payloads go to a separate interval evaluator. The planning objective therefore maximizes realized union width over every possible u_A. It is conservative under this chosen restriction, not a claim that every adaptive policy must discard legally observed outcomes.

Expand max_i Ui-min_j Lj as a maximum over four ordered pairs (i,j). Finitely many pairs allow maximization over u_A to commute with this maximum. For a pair, the known term is sum over A of (c_i-c_j)u_s. Maximizing each term on [0,1] gives max(c_i-c_j,0). The unknown-coordinate contribution is already a sum of positive upper and negative lower coefficients. Calling these terms K_ij and H_ij yields F below.

The diagonal pairs recover within-scenario unknown absolute masses. An off-diagonal pair can dominate by combining an upper endpoint under one scenario with a lower endpoint under the other. No utility sample, probability prior or learned surrogate is required. If coefficient scenarios were not a line segment, four endpoint pairs would not generally exhaust the class.

A one-query rule compares F after each eligible single addition and waits if no strict improvement exists. A visible-set rule considers currently available subsets within remaining quote cap and public

opportunities, minimizes F, prefers fewer quotes and lexicographically earlier identities, and executes only the first planned quote now. It replans at the next visible prefix. This is exact static subset planning, not a schedule over hypothetical future arrivals or a globally optimal causal policy.

$$F(A) = \max_{u_A \in [0,1]^A} W_{\text{union}}(A; u_A) \tag{8}$$

$$K_{i,j}(A) = \sum_{s \in A} \max\big(c_{i,s} - c_{j,s}, 0\big) \tag{9}$$

$$H_{i,j}(A) = \sum_{s \in A^c} \big(\max\big(c_{i,s}, 0\big) - \min\big(c_{j,s}, 0\big)\big) \tag{10}$$

$$F(A) = \max_{i,j \in \{0,1\}} \big(K_{i,j}(A) + H_{i,j}(A)\big) \tag{11}$$

Pair	Upper scenario	Lower scenario	Role
0,0	0	0	Endpoint 0 residual
0,1	0	1	Cross-scenario shift
1,0	1	0	Opposite shift
1,1	1	1	Endpoint 1 residual

Table 2: Off-diagonal pairs retain uncertainty about the continuation scenario itself.

5 S5. Analytic separation, observed zero and feasibility

The preserved analytic construction has c0=(1/4,-1/4), c1=(-1/4,1/4), both original states audited and u=(1,0). Within-scenario unknown mass is zero at both endpoints. The contrast changes continuously from 1/4 to -1/4 as lambda varies, giving union width 1/2. This proves separation without producing an observed path, a thirteenth template or a failure-rate estimate.

The saved replay says something different: all 1,620 actual fixed-panel union premiums are zero. They reuse twelve templates, aggregate prefix geometry and three panels. Their zero premiums do not refute the construction; the construction does not supply positive empirical incidence. F can still differ on a saved prefix because it considers quote outcomes other than the actually observed utilities.

The envelope supplement checks 1,104 visible subsets and 11,159 known-utility vertices. Visibility now is weaker than reachability by the past one-quote-per-checkpoint schedule. The maximum counterfactual premium 1/3 uses a subset flagged not past-reachable; the maximum among past-reachable planning subsets is 1/5. Neither is a premium of an observed actual panel. Keep these flags attached when reusing an illustrative number.

A lawful action selects a visible unaudited identity now. A static visible subset can compare current plans but cannot retroactively create enough earlier opportunities to make it already audited. Planning possibility, historically feasible audit sets and realized outcomes are three distinct objects. Confusing them turns a geometric witness into an unsupported policy gain.

$$c_0 = \left(\frac{1}{4}, -\frac{1}{4}\right), \quad c_1 = \left(-\frac{1}{4}, \frac{1}{4}\right), \quad u = (1, 0) \tag{12}$$

$$W_{\text{cond}} = 0, \quad W_{\text{union}} = \frac{1}{2}, \quad F = \frac{1}{2} \tag{13}$$

6 S6. Zero-sum L1 tubes: bound, attainment and invariant ranking

A zero-sum perturbation with L1 norm at most epsilon has equal positive and negative masses, each at most epsilon/2. With utilities in [0,1], its largest contribution is at most epsilon/2 and its smallest at least -epsilon/2. Equivalently, subtracting utility midpoint 1/2 removes the constant by zero-sum and leaves a coordinatewise magnitude bound.

For attainment with fixed audited utilities, the declared class allows two fresh unaudited coordinates, freely valued as one and zero. Put positive mass epsilon/2 on the former and negative mass on the latter for the upper shift; reverse signs for the lower shift. Leave the original coefficients at an original bound-attaining endpoint. Both fixed-scenario and scenario-union widths increase by exactly epsilon. Maximizing over audited outcomes adds the same epsilon to F.

Equality requires these coordinates, utility extremes and perturbation to be simultaneously allowed. The tube is not intersected with an independently established physically realizable continuation class. If fresh identities are forbidden, utilities cannot reach the extremes or actual occupations constrain perturbations, the upper-bound argument may survive while equality fails. If a candidate could audit the only extremal coordinates, the common-addition argument also needs rechecking.

With epsilon=2*eta*r, fixed eta and the same r for every compared candidate set, every objective receives the identical addition. Minimizers and ties remain unchanged. Enlarging a common radius can include more outcomes without changing the action. Choosing candidate-specific radii after outcomes would define a different, outcome-dependent policy and is not an evaluated repair here.

For the preserved two-state example, epsilon=1/5 changes conditioned width from zero to 1/5 and union width from 1/2 to 7/10. Positive and negative masses are each 1/10. These are arithmetic consequences of a mathematical construction, not new state-audit data.

$$\mathcal{T}_\varepsilon = \left\{ c(\lambda) + \delta : \sum_s \delta_s = 0, \sum_s |\delta_s| \le \varepsilon \right\} \tag{14}$$

$$\left|\delta^T u\right| = \left|\delta^T \left(u - \frac{1}{2}\right)\right| \le \frac{\varepsilon}{2} \tag{15}$$

$$W_{\text{tube}} = W + \varepsilon, \quad F_{\text{tube}}(A) = F(A) + \varepsilon \tag{16}$$

$$\varepsilon = 2\eta r \tag{17}$$

7 S7. The unrestricted safe interval and what it does not identify

Restricted selection cannot certify its own continuation assumption. The evaluator retains an unrestricted future suffix. K_t is the quoted signed prefix contribution. Add negative unknown prefix coefficients for the lower endpoint and positive ones for the upper endpoint. Each arm has a remaining normalized utility contribution in [0,r]; their difference is in [-r,r]. Adding it yields the safe interval below.

Safe width is the unknown prefix L1 mass plus 2r. A single available quote removes its absolute prefix coefficient. Unrestricted one-step minimax is therefore prefix_abs. Inspecting all visible identities cannot eliminate the suffix: future states may still be unseen. With arbitrarily prompt transitions to disjoint fresh identities, the worst suffix is a supremum or closure argument. A positive minimum transition delay or physical transition constraint can change attainment.

A B4/t1 construction has both arms sharing an audited utility-1/2 state through the common prefix. Compatible worlds exchange utility-one and utility-zero occupations after cost two, giving final contrasts +1/2 and -1/2. A rule reading only that common prefix cannot certify the sign in both worlds. This does not imply every online instance is unidentified or that this outer interval is sharp under a more restricted transition graph.

At the terminal checkpoint, r=0 and the fixed-path direct audit-only interval is sharp under common deterministic utility and the box model. Earlier intervals are online outer bounds. If logs, equality or utility assumptions fail, a narrow interval or positive sign cannot continue to be called certified.

$$K_t = \sum_{s \in A_t} p_s(t) u_s \tag{18}$$

$$L_t = K_t + \sum_{s \in A_t^c} \min(p_s(t), 0) - r_t \tag{19}$$

$$U_t = K_t + \sum_{s \in A_t^c} \max(p_s(t), 0) + r_t \tag{20}$$

$$W_t = U_t - L_t = \sum_{s \in A_t^c} |p_s(t)| + 2r_t \tag{21}$$

8 S8. Nine causal policies and a complementarity warning

All nine rules share identities, arrivals and quote caps. Frozen deterministic rules break score ties by ascending identity. Previously queried identities are retained, but neither their payloads nor hidden program semantics enter selection. A bound evaluator receives lawful quote values after the action is fixed. Replacing hidden future records or utility payloads while preserving the public prefix must leave actions unchanged.

One-step and set rules optimize different action spaces. A planned set contains only identities already visible and fits remaining quotes and public opportunities. Only its first member is queried; the plan can change with new arrivals. This is not an offline schedule computed from the final path. Conversely, the arrival-constrained oracle knows final coefficients and remains a lower bound, not a tenth causal competitor.

The restricted objective exhibits complementarity in a four-state construction with endpoints (1/4,-1/4,0,0) and (0,0,1/4,-1/4). Initial worst width is 1/2. Quoting the first positive-coordinate state leaves the other endpoint at 1/2; quoting the other alone also gives zero gain. Quoting both reduces the worst width to 1/4. The second quote thus gains 1/4 after the first and zero beforehand. A zero-gain stopping rule can miss a useful pair.

The construction respects zero-sum coefficients and can use remaining fraction 1/2 and active-frequency difference (-1/2,1/2,1/2,-1/2). It is a property test, not an observed common failure. It explains why an adaptive-submodularity guarantee [21] cannot be transferred without proving its required conditions for this objective and information model.

Rule	Criterion	Timing	Boundary
first_seen	Earliest arrived identity	Eager	No utility values
prefix_abs	Largest nonzero absolute prefix	Wait if none	No future commits
current_disagreement	Largest current arm difference	Wait if none	Current is not persistence
persistence_abs	Largest absolute p+r*d	Frozen rule	Diagnostic extrapolation
restricted_minimax_one	Best strict W_cond single gain	Wait at zero gain	Same public view
restricted_minimax_set	Best visible W_cond subset	Execute first, replan	No future arrivals
late_prefix	prefix_abs in last q slots	Wait beforehand	Public calendar only
union_minimax_one	Best strict F single gain	Wait at zero gain	No utility payloads
union_minimax_set	Best visible F subset	Execute first, replan	Same q and slots

Table 3: Cache quotes and planning computations are distinct costs; the oracle is not a deployable comparator.

9 S9. Checkpoint width area and the B8/q3 case

For width area J, a width observed after any quote at integer t applies to the left-step interval [t,t+1). Sum checkpoints zero through B-1 and divide by B. W_B is a separate terminal endpoint. J is neither continuous elapsed audit time nor expected recovery utility: it accounts for identification width on a stipulated recovery-cost grid.

The union-one B8/q3 widths contributing to J are 2, 7/4, 3/2, 5/4, 1, 5/6, 7/12 and 1/3. Their sum is 37/4, so J=37/32. Checkpoint six has safe width 7/12; the terminal checkpoint eight has width 1/12. Substituting one for the other is a timing error. W_8 is excluded from J because no following positive interval remains inside the horizon.

Restricted one differs only at checkpoint five: 7/8 instead of 5/6, a difference of 1/24. The union improvement is therefore 1/(24*8)=1/192. Both use checkpoints four, five and six, three quotes and final width 1/12. This is an information-allocation change in the saved replay, not a measured audit-cost saving.

Late-prefix quotes at six, seven and eight, reaches final width 1/24 and the positive unaffected certificate at eight, but has J=75/64. Relative to 37/32, area is worse by 1/64. The last quote cannot repair uncertainty accumulated earlier. Reporting only final width conceals this intermediate-information cost.

Recovery intervention work, distinct-state cache quotes, subset computations, verifier work and elapsed latency are separate columns. The horizon is not decremented by a cache quote. A deployed protocol must declare whether audits pause recovery, overlap it or consume shared resources before interpreting J as an operational objective.

$$J = \frac{1}{B}\sum_{t=0}^{B-1} W_t \tag{22}$$

$$\frac{\frac{7}{8}-\frac{5}{6}}{8} = \frac{1}{192} \tag{23}$$

$$\frac{75}{64}-\frac{37}{32} = \frac{1}{64} \tag{24}$$

Checkpoint	Restricted one	Union one	Late prefix
0	2	2	2
1	7/4	7/4	7/4
2	3/2	3/2	3/2
3	5/4	5/4	5/4
4	1	1	1
5	7/8	5/6	11/12
6	7/12	7/12	5/8

Checkpoint	Restricted one	Union one	Late prefix
7	1/3	1/3	1/3
8	1/12	1/12	1/24

Table 4: Unrestricted safe widths; t=8 is the final endpoint, not a term in J.

10 S10. All sixteen paired new-rule contracts

Every horizon/cap pairing of the new union-one and union-set rules with its same-form restricted control appears below. Delta means union minus restricted; negative area is narrower intermediate uncertainty. All final-width and quote-count differences are zero. Every pair uses the same quote checkpoints. Only the four nonzero-area pairs change at least one queried identity.

The four decreases are B8/q2 one and set, B8/q3 one and B8/q4 one, each exactly 1/192. Twelve areas tie, including every B5 pairing. No additional unaffected positive certificate appears. These results support a limited allocation effect, not a winning final-width policy.

The sixteen rows are dependent contracts over exposed paths, not sixteen independent replications or Bernoulli trials. They cannot supply a population confidence interval or a native-method leaderboard. The broader seventy-two-configuration family includes all nine policies. Its Pareto statements apply only to that family and its width/area/quote axes. The two-quote current-disagreement rule at B8/q3 remains nondominated because resources are one of those axes.

B	q	Union rule	Delta final	192 Delta area	Delta quotes	Same states
5	1	one	0	0	0	Yes
5	1	set	0	0	0	Yes
5	2	one	0	0	0	Yes
5	2	set	0	0	0	Yes
5	3	one	0	0	0	Yes
5	3	set	0	0	0	Yes
5	4	one	0	0	0	Yes
5	4	set	0	0	0	Yes
8	1	one	0	0	0	Yes
8	1	set	0	0	0	Yes
8	2	one	0	-1	0	No
8	2	set	0	-1	0	No
8	3	one	0	-1	0	No
8	3	set	0	0	0	Yes
8	4	one	0	-1	0	No
8	4	set	0	0	0	Yes

Table 5: Exact frozen v19 pairings. Query checkpoints match in all rows; cache quote counts do not count new actual audits.

11 S11. Eta sensitivity: denominator, retrospective fit and leakage

The saved aggregate views are checkpoints zero through five at B5 and zero through eight at B8: fifteen total. At the two terminal views r=0, the prefix equals the final coefficient and membership is trivial. Excluding them gives thirteen nonterminal views. That denominator includes two checkpoint-zero views, although no quote is permitted then. The eleven nonterminal query-time views are a different denominator.

At eta zero, membership tests the exact segment: every nonzero-slope coordinate must imply the same lambda in [0,1], and zero-slope coordinates must already equal the final coefficient. Equivalently, minimize L1 distance from the final vector to the segment. Its exact piecewise-linear endpoint and weighted-median breakpoints suffice for comparison with 2*eta*r. Evaluation uses the union of prefix, current and final identities; final identities remain unavailable to the selector.

Every eta row reuses the same thirteen dependent views. Inclusion grows from four to thirteen because the declared set expands. It is neither a coverage probability nor validation of a stochastic transition model. Choosing eta=1/2 because this table reaches 13/13 would inspect the outcomes needed for an independent calibration check. Repeating snapshots or adding the trivial terminal memberships cannot fix that leakage.

The grid was fixed for this diagnostic. A prospective calibration claim needs eta chosen on disjoint histories or an independently justified continuation class, frozen before future coefficients arrive and checked on reserved correction families with all violations retained. Until then, restricted objectives remain diagnostic and only unrestricted intervals certify replay comparisons.

eta	Included	Excluded	Dependent denominator
0	4	9	13
1/8	8	5	13
1/4	10	3	13
1/2	13	0	13
3/4	13	0	13
1	13	0	13

Table 6: Terminal membership is excluded; this is retrospective set compatibility, not statistical coverage.

12 S12. Verification ledger and assumption-failure checks

The scientific inputs remain 168 authored traces and five frozen support records, bound by 173 input hashes. The v19 result SHA256 is 82115c85bd66c798c897ce8827f7a270f2a3a46dbaf05ceb03aa966ba976d9d8. This v20 edit does not rerun the experiment, mutate a sealed output or generate another audit. Input, analysis and verifier snapshots remain in the v19 run.

The primary same-author separate implementation checks 540 decisions, 1,620 safe-bound cells and 1,620 union cells, plus 2,243 known-utility vertices, 5,259 lambda-breakpoint checks and 22,968 unknown-utility completions. It checks ninety memberships, seventy-two finite-family Pareto rows and fifty-six old terminal reproductions. These are computational checks, not independent experimental units.

The envelope verifier addresses sets not visited by selected policies: 1,104 visible subsets, 11,159 known-utility vertices, 6,624 tube-attainment checks and ninety common-eta ranking checks. Past-feasibility flags remain explicit. Its larger premium on an unreachable set is not presented as an implementable gain. Separate code and input locks keep this supplementary calculation distinct from primary verification.

A checksum supplies local byte agreement, not complete environment capture or external scientific review. Hidden commits invalidate the occupation partition; omitted state or stochastic behavior can invalidate common utility. Missing positive-duration observations require endpoints rather than a filled-in point estimate. Any truth outside an unrestricted interval must stop certification and trigger investigation, not be relabeled a model outlier.

Failed historical attempts remain in the preserved record. They are useful for reproduction but cannot be pooled as independent confirmation. New actual audits, worker/model calls, fits, natural histories, independent units and physical latency measurements remain zero. No real private professor consultation occurred; named methodological lenses and critic judgments are not evidence.

13 S13. Prospective study and explicit falsification criteria

This is a proposed protocol, not a completed study. Require a fixed requirement, authenticated old rejection, separately approved evaluator correction, available candidate archive and complete current-state snapshot. Preserve no-useful-reject, no-written-lesson, disputed-correction, missing-archive and failed-integration cases in the inclusion flow. Repository issues or stronger tests alone do not provide correction-history labels [17,18].

Freeze families, resource envelopes, action rules, ties and eta calibration before protected outcomes. Selectors receive matched public streams; the evaluator receives only lawfully disclosed values. Separate candidate execution, integration, lesson edits, retries, worker calls, elapsed time and failures. For stochastic states, define repeated within-state sampling and the expected-utility target before uncertainty intervals. A deterministic fixture score cannot supply that contract.

Jointly report integrated-state utility, preserved-capability loss, total work, final audit width, certificate time and any prespecified operational area. Do not switch endpoints when late-prefix wins only final width or a neutral rule wins cost. Reserve independent correction families, not repeated checkpoints, for confirmation. Set sample-size and smallest-effect rules only after obtaining an admissible sampling unit; the current 168 files are not natural histories.

Reject superiority if neutral controls retain their advantage, lineage fails to localize useful rejects, or added work outweighs the declared benefit. Reject identification if held outcomes fall outside safe intervals, state equality fails or actions read hidden futures. Reject exact tube attainment if allowed transitions cannot realize the fresh-coordinate construction. Preserve failures and nonattainment. A favorable critic cannot reverse those gates.

Gate	Falsifying observation	Response
Eligibility	Changed requirement / disputed correction	Retain flow, exclude that estimand
Information	Hidden-payload replacement changes action	Reject causal implementation
Equality	Same complete snapshot, incompatible utility	Suspend deterministic reuse
Safe bound	Held truth outside interval	Stop certificate, investigate
Tube equality	Physical class disallows extreme perturbation	Revise equality, retain justified bound
Superiority	Neutral ties benefit with lower total cost	Retain null/negative result

Table 7: Future rejection rules; no new trial or measured effect is claimed.

14 S14. Main-to-supplement routing and the preserved full record

The compact main carries the assumptions and negative results needed to assess its claim without a development chronology. S3-S4 explains the three widths; S5-S7 separates mathematical possibility, actual outcomes and safe certificates; S8-S11 makes policies and denominators inspectable. The Korean guide adds hand calculations before preserving its earlier text.

The renderer appends inherited.paper after these new sections. It is an exact object copy of the complete v19 paper, including title, subtitle, abstract, all sections, appendices, tables, figures and twenty-one reference objects. That block is labeled historical full evidence. Its chronology is not a second current main paper or an independent cohort. A v17/v19 heading there describes the source; the compact main is v20.

All reference numbers, texts, URLs and reviewed claim scopes are unchanged. Abstract-only literature-review limits remain explicit. These new sections derive our saved-evidence contract; they do not silently expand external full-text review or transfer reported external effects. STUDY_MAP.json links exact main, supplement and Korean guide headings.

Main	Supplement	Purpose
Introduction	S1, S14	Endpoints and retained history
Related Work	S1, S8, S13	Comparison roles, no guarantee transfer
Problem and Method	S2-S7	Notation and exact derivations
Experimental Setup	S8, S12	Information rights and accounting
Results	S9-S11	Arithmetic, contracts, denominator
Limitations	S12-S13	Invalidation and falsification
Conclusion	S14	Claims surviving compression

Table 8: The full prior paper follows once; no figure ID is duplicated in the new sections.

15 S15. Schedule contract and exact audited-set dynamic program

The new question fixes recovery rather than optimizing it. A schedule is a list of distinct identity quotes at strictly increasing integer checkpoints in 1..B, with at most q entries. Identity s is allowed at t exactly when its first actual appearance a(s) is no later than t. Shared SHA identities are treated as the same fixed-input utility variable across the pooled contrast; one quote reveals all three existing panel utilities. This is an inherited deterministic-cache assumption, not a newly verified property of natural agent executions.

For completed-prefix contrast p_t and queried set A, the safe interval has width W_t(A)=sum of absolute unqueried prefix coefficients plus twice the remaining fraction. Known utilities change its center, never this width. Therefore utility payloads can be removed before optimization. The executor assumes a quote returns immediately at its checkpoint without consuming recovery-clock time. All policies and oracle schedules use this same convention, but measured audit latency, failure and drift are absent.

Let D_t(A) be the smallest unnormalized left-step width sum through checkpoint t among legal schedules with queried set A after that checkpoint. Initially only A empty exists and D_0(empty)=W_0(empty)=2. A predecessor of A at t is either A itself (skip), or A without a single s that is visible by t (quote s). At t<B the transition adds W_t(A); at t=B it adds zero. The cap constrains |A|. After the final transition, divide by B and pair the value with W_B(A). Remove weakly worse pairs with at least one strict inequality.

Exactness follows by induction. At fixed t and A, future legal choices depend only on the realized arrival calendar, remaining cap and A. Future safe widths likewise depend only on the subsequent audited sets. If two histories reach this same state, replacing the larger accumulated-area history with the smaller one preserves all future actions and terminal width while weakly improving total area. Thus pruning higher-area prefixes cannot remove a Pareto point. Keeping one lexicographic witness is enough for objective recovery; separate counters retain every legal path and all equal-minimum paths.

This proof relies on fixed recovery streams, deterministic identity caching, panel-independent widths, and queries without clock feedback. Utility-aware sign-certificate timing can depend on revealed scores and is not an optimized endpoint. Query duration, changing recovery actions, stochastic utilities, nonuniform prices or state-dependent failure would require additional state variables and a different contract. No theorem of optimal causal information acquisition is claimed.

$$W_t(A) = \sum_{s \in A^c} |p_{t,s}| + 2\frac{B-t}{B} \tag{25}$$

$$J = \left(\frac{1}{B}\right) \sum_{t=0}^{B-1} W_t(A_t) \tag{26}$$

$$D_0(\varnothing) = 2, \quad D_t(A) = \min_{Q \in \mathcal{P}_t(A)} D_{t-1}(Q) + I_{t<B} W_t(A) \tag{27}$$

1. Freeze raw streams, B, cap, checkpoint slots, cache identity and the safe-width formula.
2. Project raw rows to history/arm/budget and commit cost/identity fields only.
3. Reconstruct prefix coefficients and cumulative visible identities at every integer checkpoint.
4. For each checkpoint and audited set, extend skip and every legal single-identity quote.
5. Retain minimum area prefix per queried set, a stable witness, and separate total/tie counts.
6. At B compute terminal width, normalize area and remove dominated objective pairs.
7. Compare every saved policy schedule with the frontier; do not pass the oracle schedule to an online policy.

Algorithm 1: Offline structural optimization; full future structure is privileged and utility payloads are absent.

16 S16. Eight complete frontiers and the limits of a singleton result

The independent enumerator exhausted 33,658 schedule-contract combinations across eight settings. Caps are nested: a one-quote schedule occurs again when cap is two, three or four. Therefore 33,658 is neither a count of unique schedules nor an independent sample size. The maximum-cap universes contain 1,175 schedules at B5 and 24,729 at B8, or 25,904 distinct horizon/schedule pairs overall. They all reuse twelve authored templates and nine identities.

Every observed frontier is a singleton objective point. This means some legal schedule jointly attains the smallest terminal width and smallest area for that setting. It does not mean there is a unique schedule: equal-objective witness counts range from two to forty. Nor does it imply that every future arrival/occupation process has a singleton frontier. The DP is exact over this finite contract; singleton behavior is a result of these particular exposed streams.

The horizon-five area floor is 6/5 once cap is at least two. With two appropriate identities already queried, all completed-prefix contributions that could widen the intermediate interval are removed. Remaining suffix radius still prevents a zero area. More quotes can improve the terminal width without further changing J. At B8, the optimal extra identities beyond two arrive at checkpoints four and five; their early legal purchase reduces both the final width and the later checkpoint areas.

Pareto dominance here has two axes at a fixed allowance. The older nine-rule table additionally minimized quotes actually spent. A policy spending fewer than q may be nondominated on that older three-axis restricted-family table yet dominated here, since this study permits using any schedule up to the same cap. These are different comparison questions, not an inconsistency in the old results.

B / cap	Legal schedules	Terminal width	Area J	Equal-objective schedules	Witness checkpoints
5 / 1	25	1/3	31/25	2	2
5 / 2	207	1/5	6/5	4	1,2
5 / 3	715	2/15	6/5	28	1,2,3
5 / 4	1175	1/15	6/5	40	1,2,3,4
8 / 1	45	1/8	75/64	12	1
8 / 2	753	1/12	37/32	36	1,2
8 / 3	6009	1/24	73/64	24	1,2,4
8 / 4	24729	0	217/192	30	1,2,4,5

Table 9: Complete schedule universes on fixed authored streams. Nested-cap counts repeat schedules; every oracle reads future structure but no utility values.

17 S17. B8 three-quote trace: structural lookahead removes the family tradeoff

The old comparison was a genuine timing reversal within declared rules: FIFO and union-set had final width 1/12 and area 37/32, whereas late-prefix had final width 1/24 and area 75/64. The complete schedule search finds a witness with final width 1/24 and area 73/64. Its queries occur at checkpoints

one, two and four, rather than late-prefix's six, seven and eight. These arrival-feasible choices exploit knowledge of which early identities will matter after later cancellations.

The witness first quotes SHA a8cd6336a49059c9bfc450b7174014a1eed4faac438cbdf1493d3b02742b4ed9 at t1, then aa8bd06ef866ce5ccd80ef8bb80963031c91a54dfc75c083825e4876b0b9aec8 at t2, then aa6d20fb002042976aeb34c983d92fda86030eb051a30e976ec78c11161de156 at t4. The first two identities arrived at zero; the last arrived at four. Choosing that last identity from all arrivals at four uses the realized future occupation path in optimization. Legal availability is weaker than causal selectability without advance knowledge.

Its width curve from t0 through t8 is (2,7/4,3/2,5/4,1,19/24,13/24,7/24,1/24). The first five widths sum to 15/2. Adding the t5..7 widths gives 73/8; division by eight gives J=73/64. The endpoint 1/24 is not included in that area. FIFO's corresponding late widths are 5/6,7/12,1/3; they exceed the witness by 1/24 each, so the area gap is (3/24)/8=1/64. Late-prefix's area gap is 75/64-73/64=1/32.

This trace does not establish that an auditor could predict which identity to buy. It demonstrates where the declaration of nine rules left structural headroom. A prospective method must select using only current prefix information and be evaluated on new histories; encoding this hindsight schedule into a rule and reusing the same exposed streams would not establish such a method. Sign-certificate timing was not optimized, and the old unreached certificates remain unchanged in the comparison ledger.

B8/q3 object	Final width	Area	Area gap	Terminal gap	Future rights
Structural witness	1/24	73/64	0	0	Full arrivals/occupation
FIFO	1/12	37/32	1/64	1/24	Prefix only
Union set	1/12	37/32	1/64	1/24	Prefix only
Restricted one	1/12	223/192	1/48	1/24	Prefix only
Late prefix	1/24	75/64	1/32	0	Prefix + public slots

Table 10: Gaps are to the same fixed-cap structural point. No forecastability or natural recovery improvement is inferred.

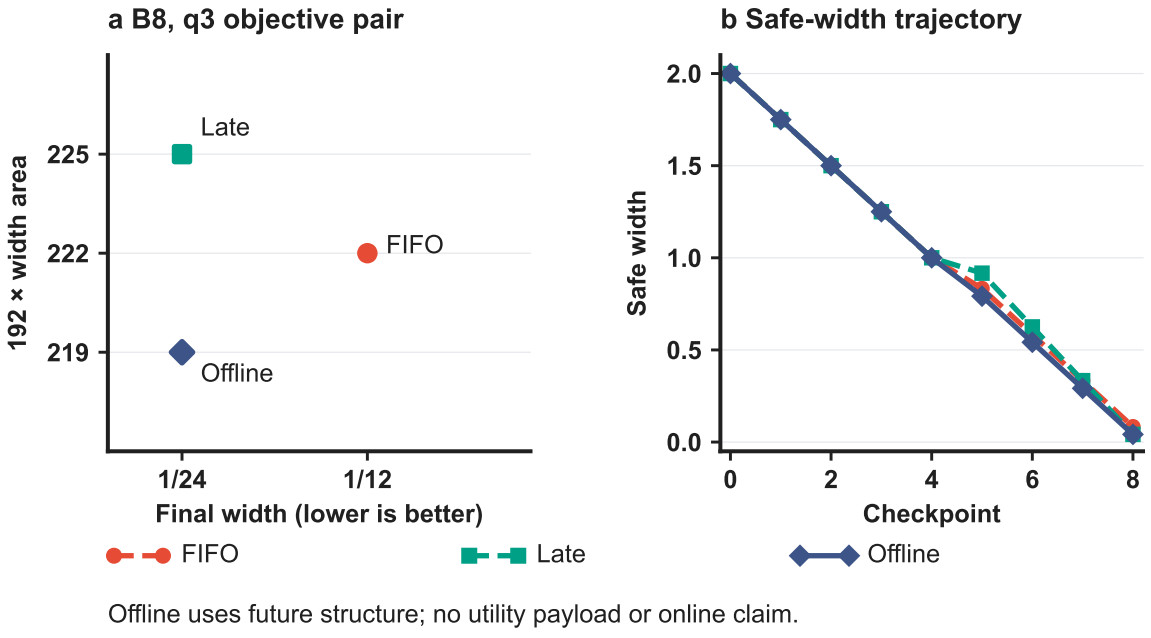


Figure 1: B8/q3 complete-schedule structural frontier versus FIFO and late-prefix: the oracle attains (1/24,73/64) and reads future arrivals/occupations. Legal query timing does not make this a causal policy; utility payloads are absent and no natural recovery gain is inferred.

18 S18. Complete policy comparison and verification ledger

All 72 preserved policy schedules are legal under the new contract and reproduce their original terminal width, area, quote use and certificate records exactly. Twenty-eight lie on the complete two-axis frontier; 44 are dominated. Since each contract's frontier is a singleton, the endpoint-

conditioned area and area-conditioned endpoint gaps reduce to the differences from that point. The implementation still computes the constrained definitions explicitly rather than assuming singleton behavior.

The analysis freezes 168 authored trace snapshots and four v19 support/result files, totaling 172 input hashes. The independently written verifier imports no analysis function: it reconstructs coefficients by summing integer unit occupations and enumerates every schedule rather than applying DP. It matches all frontier points, equal-objective multiplicities, per-subset minimum areas, schedule counts and witnesses; all fourteen malformed or impossible schedules are rejected. The two implementations have the same author and inputs, so this is alternative arithmetic verification, not external replication or proof of complete natural-state logging.

The structural projection contains only history, recovery arm, budget, and commit cost/identity. Replacing all utility and panel payloads leaves this projection unchanged. State-value consistency and [0,1] ranges are checked separately as inherited admission assumptions. No utility-dependent objective is sent into the DP. Hash agreement supports provenance; it does not create new evidence of deterministic re-audits, causal recovery effect, statistical confidence coverage or operational latency savings.

The canonical files are runs/research_advancement_v21_20261019_01/second_chance/analysis_01/results.json, frontiers.json, policy_comparisons.json and case_traces.json. The separate verifier receipt is verification_01/verification.json beside analysis_01. PLAN.json and input_lock.json bind code and snapshots before new outcomes. Read original unnormalized schedule counts alongside caps; do not treat rows or checkpoint repetitions as population samples.

B/q	Policy	Final W	Area J	Gap W / J	On frontier
5/1	first_seen	7/15	32/25	2/15 / 1/25	No
5/1	prefix_abs	1/3	31/25	0 / 0	Yes
5/1	current_disagreement	1/3	31/25	0 / 0	Yes
5/1	persistence_abs	1/3	31/25	0 / 0	Yes
5/1	restricted_minimax_one	1/3	31/25	0 / 0	Yes
5/1	restricted_minimax_set	1/3	31/25	0 / 0	Yes
5/1	late_prefix	1/3	32/25	0 / 1/25	No
5/1	union_minimax_one	1/3	31/25	0 / 0	Yes
5/1	union_minimax_set	1/3	31/25	0 / 0	Yes
5/2	first_seen	1/3	31/25	2/15 / 1/25	No
5/2	prefix_abs	1/5	91/75	0 / 1/75	No
5/2	current_disagreement	1/5	6/5	0 / 0	Yes
5/2	persistence_abs	1/5	6/5	0 / 0	Yes
5/2	restricted_minimax_one	1/5	6/5	0 / 0	Yes
5/2	restricted_minimax_set	1/5	6/5	0 / 0	Yes
5/2	late_prefix	1/5	94/75	0 / 4/75	No
5/2	union_minimax_one	1/5	6/5	0 / 0	Yes
5/2	union_minimax_set	1/5	6/5	0 / 0	Yes
5/3	first_seen	2/15	6/5	0 / 0	Yes
5/3	prefix_abs	2/15	91/75	0 / 1/75	No
5/3	current_disagreement	2/15	6/5	0 / 0	Yes
5/3	persistence_abs	2/15	6/5	0 / 0	Yes
5/3	restricted_minimax_one	2/15	6/5	0 / 0	Yes
5/3	restricted_minimax_set	2/15	6/5	0 / 0	Yes
5/3	late_prefix	2/15	91/75	0 / 1/75	No
5/3	union_minimax_one	2/15	6/5	0 / 0	Yes
5/3	union_minimax_set	2/15	6/5	0 / 0	Yes
5/4	first_seen	2/15	6/5	1/15 / 0	No
5/4	prefix_abs	2/15	91/75	1/15 / 1/75	No
5/4	current_disagreement	1/15	6/5	0 / 0	Yes
5/4	persistence_abs	1/15	6/5	0 / 0	Yes
5/4	restricted_minimax_one	1/15	6/5	0 / 0	Yes
5/4	restricted_minimax_set	1/15	6/5	0 / 0	Yes
5/4	late_prefix	2/15	91/75	1/15 / 1/75	No
5/4	union_minimax_one	1/15	6/5	0 / 0	Yes
5/4	union_minimax_set	1/15	6/5	0 / 0	Yes

B/q	Policy	Final W	Area J	Gap W / J	On frontier
8/1	first_seen	1/8	75/64	0 / 0	Yes
8/1	prefix_abs	1/6	227/192	1/24 / 1/96	No
8/1	current_disagreement	1/6	227/192	1/24 / 1/96	No
8/1	persistence_abs	1/6	227/192	1/24 / 1/96	No
8/1	restricted_minimax_one	1/6	227/192	1/24 / 1/96	No
8/1	restricted_minimax_set	1/6	227/192	1/24 / 1/96	No
8/1	late_prefix	1/8	19/16	0 / 1/64	No
8/1	union_minimax_one	1/6	227/192	1/24 / 1/96	No
8/1	union_minimax_set	1/6	227/192	1/24 / 1/96	No
8/2	first_seen	1/12	37/32	0 / 0	Yes
8/2	prefix_abs	1/8	75/64	1/24 / 1/64	No
8/2	current_disagreement	1/8	75/64	1/24 / 1/64	No
8/2	persistence_abs	1/8	75/64	1/24 / 1/64	No
8/2	restricted_minimax_one	1/8	75/64	1/24 / 1/64	No
8/2	restricted_minimax_set	1/8	75/64	1/24 / 1/64	No
8/2	late_prefix	1/12	227/192	0 / 5/192	No
8/2	union_minimax_one	1/8	7/6	1/24 / 1/96	No
8/2	union_minimax_set	1/8	7/6	1/24 / 1/96	No
8/3	first_seen	1/12	37/32	1/24 / 1/64	No
8/3	prefix_abs	1/12	7/6	1/24 / 5/192	No
8/3	current_disagreement	1/8	75/64	1/12 / 1/32	No
8/3	persistence_abs	1/12	223/192	1/24 / 1/48	No
8/3	restricted_minimax_one	1/12	223/192	1/24 / 1/48	No
8/3	restricted_minimax_set	1/12	37/32	1/24 / 1/64	No
8/3	late_prefix	1/24	75/64	0 / 1/32	No
8/3	union_minimax_one	1/12	37/32	1/24 / 1/64	No
8/3	union_minimax_set	1/12	37/32	1/24 / 1/64	No
8/4	first_seen	1/12	221/192	1/12 / 1/48	No
8/4	prefix_abs	1/24	7/6	1/24 / 7/192	No
8/4	current_disagreement	1/8	75/64	1/8 / 1/24	No
8/4	persistence_abs	1/24	37/32	1/24 / 5/192	No
8/4	restricted_minimax_one	1/24	37/32	1/24 / 5/192	No
8/4	restricted_minimax_set	1/24	221/192	1/24 / 1/48	No
8/4	late_prefix	1/24	7/6	1/24 / 7/192	No
8/4	union_minimax_one	1/24	221/192	1/24 / 1/48	No
8/4	union_minimax_set	1/24	221/192	1/24 / 1/48	No

Table 11: All nine policies in all eight settings; two axes at a fixed cap. Existing nonattainment and certificate timing remain in the machine-readable comparison rows.

19 S19. Primary trajectory resources and an unclosed transfer gate

Three additional primary resources were directly inspected on 2026-10-19 and are recorded in literature/RESOURCES.json with selected-passage hashes. They are reading and prospective-design resources, not new numbered references or executed baselines. Existing twenty-one references remain unchanged. Only bounded passages are retained; no external PDF, dataset download, model inference or remote code execution was performed.

AgentDebug and AgentErrorBench describe annotated failure trajectories and targeted reruns. The version-one algorithm labels its second stage as LLM detection without counterfactual rollout, while nearby prose describes counterfactual testing; that ambiguity prevents treating the read paper alone as a verified executable intervention protocol. Who&When instead supplies agent/step attribution labels with distinct reference-answer access settings. AgentDebugX documents trace capture and rerun branches; restoring a selected event requires runner checkpoint support. These observations do not establish our complete deterministic-state condition.

Paper URLs: https://arxiv.org/html/2509.25370v1 ; https://proceedings.mlr.press/v267/zhang25cq.html and https://arxiv.org/html/2505.00212v3 ; https://arxiv.org/html/2607.18754v1 . Official code/data links and review locators are in the resource records. Who&When's public Hugging Face viewer exposes trajectory and attribution fields; its data license was not established by the inspected card. AgentErrorBench's Drive link resolved but its file inventory and data license were not verified.

Repository MIT notices are software notices and do not by themselves certify underlying environment-data permissions.

Our proposed next stage is to select a permitted natural trajectory family, capture complete prompts, persistent memory, tool/environment versions, task inputs and relevant hidden state, and test identical-state replay before defining an identity cache. Preserve pre/post correction branches and a fixed later-input panel. Audit allocation must be fixed without final occupations and use measured calls, tokens, failures and elapsed query time. Evaluate integrated-state recovery and collateral retention separately from correct error attribution. None of the inspected resources automatically supplies that entire contract; this transfer gate remains open.

20 Complete preserved v19 scientific record

The following complete record preserves the earlier scientific sections, negative results, equations, tables, algorithms, figures and appendices. Its development-stage statements retain their original scope. The current compact conclusions and detailed derivations precede this archived material; retaining an earlier result does not turn it into an independent replication.

SecondChance: Recovery Audits with Missing State Observations: Conditional minimax, model misspecification and the price of waiting in prefix-only recovery audits

Neutral working paper v19, 2026-10-19. New finite saved-evidence diagnostics; no new independent data, real model/API calls, native-method trial or physical acquisition. Previous v17 evidence and v18 presentation are retained.

Correcting an evaluator does not establish recovery of the current agent state. SecondChance separates historical rescoring, integrated-state audits and the information available to an audit selector. Earlier authored experiments and partial-observation diagnostics retain twelve histories, 168 policy paths and nine complete state identities. V16 introduced three selectors without future commits. V17 asks what a robust selector actually protects when the continuation model can be wrong. We prove that one-query minimax under unrestricted future states reduces to the existing absolute-prefix rule, then implement a restricted persistence/cancellation segment with one-query and visible-set minimax allocation. Alongside FIFO, current-state disagreement, persistence ranking and a last-opportunity prefix control, seven policies yield 56 dependent configurations, 420 decisions and 1, 260 unrestricted safe bounds. The restricted segment excludes the actual final coefficient vector in nine of thirteen nonterminal views. At horizon eight and quote cap three, late-prefix reaches width 1/24 and a positive certificate at checkpoint eight; restricted minimax reaches 1/12 and no certificate. Waiting increases normalized checkpoint uncertainty area from 37/32 for restricted-set to75/64. A four-state construction shows zero individual minimax gains but a positive two-query gain, so a diminishing-returns assumption cannot be silently transferred. All safe bounds use the unrestricted future suffix, irrespective of selection model. Exact verification preserves the prior negative evidence and distinguishes conditional robustness, final width, certificate delay and cached quote cost. Actual prospective audits and natural histories added remain zero; no population recovery or scheduling superiority is claimed. The continuation separates unknown-scenario objectives and shows no final-width gain in16 paired contracts; uniform L1 inflation changes model inclusion but not candidate rankings.

- A state-bound recovery contract separating saved-answer rescoring, fresh candidate execution and fresh integrated-state audits, with strong neutral controls and all-template denominators.
- An executed controlled-censoring study on nine complete state identities across 168 exposed authored paths: 512 audit sets per budget/panel and 3,072 exactly checked contrast bounds.
- Sharp audit-only contrast endpoints that preserve shared-state dependence, plus retrospective minimum audit certificates and explicit counterexamples when complete-log or deterministic-state assumptions are removed.
- A reproducible continuation of the v11/v12 negative results: fresh committed-state observations, two signs of sampling error, neutral ties/advantages, preserved failed attempts and a staged natural-model confirmation protocol.
- A predeclared utility-blind coefficient schedule achieves the minimum fixed-cardinality contrast width and separates exact-width optimality from outcome-dependent sign stopping and retrospective certificates.
- A prefix-only audit interface, unknown-future suffix bound and causal replay with exact arrival-feasible oracle comparison; counterexamples prevent transferring offline greedy optimality to an online setting.

- A conditional robust-allocation benchmark that separates worst scenario-conditioned width from unknown-scenario identification, tests continuation misspecification, exposes query complementarity and compares delayed cheap audits with their uncertainty-time cost.
- The continuation distinguishes the worst width conditional on a fixed scenario, the union of identification intervals when the scenario itself is unknown, and a pre-outcome worst union objective used for utility-blind audit selection. The last has four endpoint-pair cases; a two-state analytic construction separates zero conditioned width from union width1/2. This mathematical construction must not be substituted for observed incidence.

21 Preserved v19: Introduction

An evaluator patch can reverse yesterday's rejection while leaving today's system broken. Consider a correct signed-arithmetic implementation that was rejected because an evaluator compared its answer against an absolute value. The incumbent remains the wrong program; a later, valid output-format fix may already have been applied; and a lesson written from the rejection may still recommend the wrong procedure. Rescoring the archived answer changes the historical verdict. It does not execute an integration, preserve the later fix or withdraw the lesson. A deployment decision therefore requires a behavioral endpoint on the actual current state, rather than a corrected score alone.

This distinction matters for systems that update persistent textual state. SkillOpt gates bounded skill edits on validation and retains rejected edits as feedback [1]; Reflexion conditions future trials on feedback-derived reflective memory [2]. Self-Refine and CRITIC improve outputs through different feedback loops [3,4], while SCoRe learns correction behavior through online reinforcement learning [6]. These mechanisms motivate inspecting decisions and memory after an evaluator correction. They do not by themselves answer the narrower identification question studied here: after the requirement remains fixed and its evaluator is corrected, what evidence establishes that an archived alternative has actually repaired the integrated current state?

We make that question executable before testing it on language models. SecondChance is a small deterministic research instrument with an immutable requirement, authored faulty/corrected evaluators, a fixed archive, logged state commits and fresh state audits. It separates E1, rescoring the same saved output; E2, freshly invoking a candidate; and E3, auditing the integrated artifact with its surviving lesson and valid later changes. Figure 1 depicts these endpoints. The construction permits exact inspection and falsification of a measurement contract, but supplies no estimate of natural evaluator faults, LLM recovery probability or method superiority.

We ask three linked questions. RQ1: can saved-score and fresh-candidate success disagree with integrated-state recovery? RQ2: does lineage-first joint repair beat a strong neutral early-stop procedure under the same authored caps? RQ3: when recovery is summarized over intervention cost, how much can a coarse observation grid hide? Twelve authored histories yield 168 nested policy-budget paths. The neutral procedure ties terminal recovery and is better on cost and collateral loss at budget five. A follow-up freezes every path and makes 292 new state audits. It reveals both missed recovery and understated collateral degradation, with discrepancies covered by a conditional commit-geometry bound. Thus the paper's contribution is a falsifiable recovery measurement and control protocol, including negative policy evidence, rather than a demonstrated new LLM recovery algorithm.

Incomplete observation raises a further decision: which policy comparisons remain warranted when some state audits are unavailable? Version 14 addresses this question by deliberately deleting saved observations from the exposed deterministic paths. It treats unknown utilities as unknown, retains equality for the same complete state and fixed input, and bounds the policy contrast directly. An individual area can remain uncertain even when common uncertainty cancels from a paired comparison. This is a new finite diagnostic, not a new agent recovery experiment.

The v14 minimum audit subsets were selected retrospectively using the complete outcomes. That does not supply a deployable order. V15 instead constructs an order using complete logged state identities and occupation coefficients before revealing panel utilities. The distinction is narrow but consequential: a width-optimal known-path schedule can be implemented without utility leakage, yet it still need not match the cheapest outcome-specific sign certificate or operate before future commits are known.

V17 makes the robustness assumption itself inspectable. An unrestricted one-step minimax criterion reduces to prefix_abs; a restricted persistence/cancellation model changes decisions but misses many actual continuations. A cheap rule that waits for the last public query opportunities can narrow the terminal interval while carrying greater intermediate uncertainty. Figure 8 separates these outcomes from real prospective audit savings.



Figure 2: Three endpoints after the same evaluator correction. E1 changes a saved verdict; E2 invokes an archived candidate; E3 newly executes the integrated state. At budget eight, selection arms have 12/12 development-positive candidates but 4/12 initial-state recoveries, while rescoring alone restores 0/12. Counts describe twelve authored histories, not language-model performance.

22 Preserved v19: Related Work

We organize comparisons by the decision each method changes, the feedback it may access and the object it evaluates. This avoids treating every reflection or verification method as an interchangeable repair baseline. Our twenty references support particular design choices; their reported benchmark numbers are not pooled with our executor results.

22.1 Preserved v19: Correction, selection and persistent memory

Self-Refine uses one model to generate, critique and refine an output [3]; CRITIC incorporates tool interaction when validating and revising an output [4]. Huang et al. examine intrinsic reasoning self-correction without external feedback [5]. Their scope is narrower than all correction, especially in light of later learned correction strategies such as SCoRe [6]. Our evaluator correction is an explicit external event, and our current study runs no language model. These papers motivate controlling feedback and update rights; neither their successes nor their limitations can be imported as our empirical result.

Verifier-based selection ranks generated candidate solutions [7]. SkillOpt instead optimizes a persistent skill document through bounded edits and a validation gate, with rejected-edit feedback retained for later updates [1]. Consequently, retaining rejected candidates is not our novelty claim. We audit what a changed evaluator requires after those decisions: immutable old outputs, a common current-state fork, actual integration, preserved later changes and a fresh integrated-state outcome. The fixture is a fixed-pool experiment; generating a new corrective skill remains a separate comparison.

Reflexion provides the closest motivation for examining feedback-derived lessons [2]. Zep maintains temporally scoped facts and invalidates inconsistent edges when information changes [8]. Our setting holds the requirement fixed and corrects its erroneous implementation; world-state change and evaluator correction have different approval conditions. We therefore test a same-artifact retain/suspend lesson fork and leave native Reflexion, Zep and SkillOpt reproduction unmeasured.

Family	Object changed	Feedback contract	Comparison here
Self-Refine / CRITIC [3,4]	Current response	Self feedback / tool-derived critique	Design context; not run
Intrinsic correction / SCoRe [5,6]	Reasoning response / learned correction policy	No external feedback / online training	Different training and feedback rights
Verifiers / SkillOpt [7,1]	Candidate choice / persistent skill	Verifier ranking / validation-gated edits	Fixed pool and correction eligibility specified
Reflexion / Zep [2,8]	Reflective text / temporal fact graph	Task feedback / changing information	Same-artifact lesson fork; native results NR
Neutral early-stop	Integrated fixture state	Same corrected development grader	Executed strong control

Table 12: Mechanism comparison, not an external performance leaderboard. NR means not reported because the method was not executed.

22.2 Preserved v19: Evaluation validity and executable correctness

Proxy optimization can improve a development score without improving a separately evaluated objective [9]. Model-selection bias and adaptive holdout reuse make evaluation access part of the method, rather than an implementation detail [10,11]. These motivate freezing candidate pools, audit inputs and disclosure boundaries. Our researcher can inspect every fixture label; controller non-use is documented but is not independent adjudication or a secret final test.

EvalPlus shows why finite code tests can miss incorrect implementations [17]. SWE-bench supplies repository issue-resolution tasks that could support a richer future environment [18], but neither resource automatically labels an old evaluator as wrong under an unchanged requirement. Such labels, the historical rejected edits and their downstream lessons must be collected separately. ReAct's interleaving of reasoning and environment actions also makes clear why a prompt or program hash alone may omit relevant agent state [19].

22.3 Preserved v19: Resource curves and reproducible comparisons

Henderson et al. emphasize nondeterminism and reporting variability [12]; Agarwal et al. address uncertainty in aggregate few-run comparisons [13]. We retain history-level denominators and avoid confidence intervals that would falsely turn designed templates into sampled natural histories. Local process counts and wall durations are execution resources, not additional independent trials.

Performance profiles compare distributions of a declared performance metric [14], and COCO makes problem instances, targets and function-call runtime explicit [15,16]. Those ideas motivate reporting both terminal behavior and an intervention-cost curve. Our integrated utility can decrease after a repair, unlike a retained best-so-far objective; hitting a target once does not guarantee its subsequent persistence. The area and geometry bound below are our conditional measurement argument, not a theorem imported from numerical-optimization benchmarks.

Partial identification concerns the values compatible with observed information and assumptions; sampling uncertainty is a separate issue. The distinction between an identification region and confidence coverage is explicit in the primary abstract of Imbens and Manski [20]. Our fixed authored cohort supplies no sampling-based population confidence interval, and we do not implement their inferential procedures.

23 Preserved v19: Problem Formulation

The unit of comparison is a frozen history under a verified correction to a fixed requirement. The state includes the executable artifact and any active lesson that affects routing. Correction eligibility, selection success and current recovery are separate predicates.

23.1 Preserved v19: Correction eligibility and behavioral endpoints

The unchanged requirement G is exact signed addition of a/b and c/d. The response grammar permits a bounded signed integer, decimal or nonzero-denominator fraction after FINAL=. Numerically equivalent rational forms are accepted, so FINAL=-2/3 and FINAL=-4/6 denote the same value. Parsing uses a bounded grammar and Python Fraction; response text is never evaluated as code. Non-numeric values, zero denominators, excessive numeric lengths and arbitrary prefixes are rejected. Approximate decimal values are not silently accepted as exact rationals.

The authored defect replaces the exact signed target with its absolute value. Under the negative development sum, an absolute-value program is accepted while an exact signed program is rejected. The accepted correction removes that absolute-value operation while preserving the task and valid-answer contract. Old outputs and old decisions are appended to, never replaced by corrected scores. A change in requested sign, task identity, accepted alternatives or evaluation objective would instead define a new task or a disputed correction and must not be counted as the same correction event.

Correction eligibility in this fixture follows from the authored arithmetic contract, not independent human review. In an empirical scientific-label setting, the fixed requirement, counterexample, old assertion, corrected assertion and valid alternatives would need a separate adjudication procedure. A model critic agreeing with the correction would not provide that independence. Ambiguous annotations, requirements changes and unsupported corrections should remain pending rather than become useful-rejection evidence.

Endpoint E1 is V_new applied to the exact same saved response bytes; it measures historical score change. Endpoint E2 is an actual fresh candidate invocation on permitted development data; it measures what that candidate newly does under its own state. Endpoint E3 is a fresh audit of the integrated current-state checkpoint, including retained lessons and routing. E2 can be positive for

a development-memorizing candidate while E3 fails on a different input. E1 has no state-transition semantics at all.

$$G(x) = \frac{a}{b} + \frac{c}{d} \tag{28}$$

$$V_{\text{old}}(y, x) = I(\text{parse}(y) = |G(x)|) \tag{29}$$

$$V_{\text{new}}(y, x) = I(\text{parse}(y) = G(x)) \tag{30}$$

$$\Delta_{\text{saved}} = V_{\text{new}}(y_{\text{saved}}, x) - V_{\text{old}}(y_{\text{saved}}, x) \tag{31}$$

23.2 Preserved v19: Complete state and decision lineage

A checkpoint contains a program operation, the lesson text and its active/absent/suspended status, and a valid later change that must survive repair. The latter fixes the FINAL= output prefix and the JSON trace format. Candidate artifacts encode exact rational addition, absolute-value addition, development-input memorization, integer truncation or a constant zero response. Each candidate runs in a separate local Python process. Its request bytes, response bytes, state hash, worker hash, return code and observed process duration are recorded.

The history records the candidate archive, saved outputs, old accept/reject events, incumbent and any actually written fixture lesson. The fixed authored lesson writer can record the instruction to fall back to the absolute-value solver for opposite-sign addition. This is not a model reflection. The initial audit uses negative same-sign terms; the later panel uses opposite-sign terms with a negative total. The constructed lesson can therefore preserve initial success after integration yet break success on the later input. All those task values and the writer rule are visible to the researcher.

Lineage is also explicitly authored. An aligned hint points to the useful exact candidate; a misleading hint points to the development-memorizing candidate. Every policy receives the same hint and archive. The neutral policy chooses not to use the hint for its ordering. This design exposes the consequences of reliable versus unreliable localization; it does not learn a localization model or estimate the prevalence of good historical lineage.

A saved rescore and a candidate invocation do not themselves mutate the incumbent. Only the logged integration and lesson-suspension events can change this fixture state. Integration verifies the candidate artifact and preserved valid-change hash. A common parent hash binds all policy forks. The pre-lesson checkpoint and final checkpoint are retained separately. This complete state definition is a testable assumption, not a property guaranteed for general agents with filesystem state, latent memory, tools or external services.

Freeze the requirement, faulty/corrected assertions, archive bytes, old decisions, incumbent and written lesson.
Fork the same initial state for each already declared policy and budget.
Charge each rescore, fresh candidate execution, integration and lesson action; retain unused allowance.
Commit state only after integration or lesson treatment and store the state/parent/artifact hashes.
Audit isolated state copies on the fixed panels; keep audit responses outside policy selection.
Retain no-useful-candidate, no-written-lesson, no-gain, invalid, timeout and missing-audit outcomes.

Algorithm 2: Research instrument, not a model-driven autonomous repair algorithm.

24 Preserved v19: Method: A Recovery Audit Contract

The instrument runs matched intervention procedures and observes state-bound outcomes. It does not use a critic's agreement as evidence of correctness. A policy result is identified only relative to a recorded state, allowed information, intervention budget, audit panel and failure/missingness rule.

24.1 Preserved v19: Endpoint denominators and the lesson contrast

A current initial recovery requires success on the fresh initial-input audit of the integrated state. The main all-started burden counts no initial recovery or failure on the later-input panel. It keeps templates without useful rejected candidates in the denominator. Conditional persistence failure divides the number of initial successes that fail later by the number of initial successes; zero initial successes yield null, not zero. Joint initial-success/later-failure frequency and total later failure are separate all-template quantities.

The label later refers to a different input executed against the same final checkpoint. No further agent evolution, model training, new selection or calendar-time drift occurs between the initial and later panels. Thus the stored strict_recurrence field denotes cross-panel maintenance failure in the fixture. It is not a measured long-term evolutionary recurrence rate. A future recurrence study must specify the subsequent evolution process and preserve its additional decisions, exposure and resources.

Unaffected-task performance is paired before and after intervention. Absolute failure after repair and incremental loss from the same baseline are different quantities. This matters because selecting a development-memorizing program can destroy an unaffected positive-addition capability even when the corrected development score is perfect. Templates with such collateral losses remain in every relevant table.

The fixed-artifact lesson contrast is built only after selection and integration have happened once. The exact pre-lesson bytes are copied into retain and suspend branches, preserving program artifact, valid later changes, routing and allowance. Selection is not rerun. This controls the integration mediator that would otherwise differ between whole procedures. In a general model experiment, treatment assignment, context length, allowed correction knowledge and memory retrieval would also need explicit control. Here the mechanism is visibly authored in the worker, so its contrast verifies implementation rather than discovering an LLM memory mechanism.

$$B_{\text{all}} = \left(\frac{1}{N}\right) \sum_{h=1}^{N} I(R_h = 0 \text{ or } P_h = 0) \tag{32}$$

$$F_{\text{cond}} = \frac{\sum_h I(R_h = 1 \text{ and } P_h = 0)}{\sum_h I(R_h = 1)} \tag{33}$$

$$\Gamma = Y_{11} - Y_{10} - Y_{01} + Y_{00} \tag{34}$$

24.2 Preserved v19: Cost-indexed recovery utility

Version 11 observes the current state at policy-cost coordinates 0, B/4, B/2, 3B/4 and B and computes a normalized left-step area. A terminal success can therefore have zero recorded area. For example, an initially failing state integrated successfully at cost four under cap five is still failing at the last left endpoint 3.75 and successful only at the final observation five. The resulting zero is correct for the declared coarse statistic, but it is not the area of the complete committed-state path.

Let c_i be every committed-state cost, including zero and a terminal boundary B. Let s_i be the state in force after c_i, held until the next commit. If the log contains every state mutation, the worker is a pure deterministic function of that state and the fixed audit input, and each positive-duration state has actually been freshly audited, then the state-path area is identified by the observed step sum. This is a conditional statement about this executor and panel. It is not a claim that one output identifies a stochastic agent's expected capability.

The argument is direct: partition the cost axis at every complete state commit. The executor and input do not change within each resulting interval, so its binary output is constant there. A fresh observed value for that state supplies the interval's utility, and integration over the finite partition gives A_C. Audits are not inferred from saved-score positives or candidate identifiers. A missing positive-duration audit leaves its area contribution unknown in [0,width/B]. A missing state that exists only at the endpoint B can leave the area defined while the terminal score remains missing.

The cost axis excludes the common external audit envelope. Adding audit spending retroactively to the controller's five- or eight-unit axis would change the estimand and possibly the feasible repair procedure. We instead report actual audit subprocess counts and elapsed time as a separate resource vector. These are local process measurements, not model tokens, billed dollars or a paired hardware-speed benchmark.

$$A_G = \left(\frac{1}{B}\right) \sum_{j=0}^{J-1} (g_{j+1} - g_j) U(s(g_j)) \tag{35}$$

$$A_C = \left(\frac{1}{B}\right) \sum_{i=0}^{K-1} (c_{i+1} - c_i) U(s_i) \tag{36}$$

$$\Delta_A = A_C - A_G \tag{37}$$

24.3 Preserved v19: Commit reconstruction and an error bound

For each coarse cell [g_j,g_(j+1)), let t_j be its first strictly interior state-changing commit. If there is no such commit, set t_j to the right boundary. A commit at a coarse grid point is already observed there, and a metadata event leaving the state bytes unchanged creates no new exposure interval. Until t_j the state equals the coarse left-endpoint state. Afterwards each bounded utility can differ from that left value by at most one. Summing those potentially exposed lengths proves the bound below without a monotonicity assumption.

The bound is sharp in a one-change cell: a utility jump from zero to one, or from one to zero, at the first interior commit persists through the cell. For the budget-five integration at four, the exposed interval is [4,5), yielding E=1/5. A coarse initial-area value of zero and a freshly observed commit-aware value of 1/5 attain the bound. Grid-aligned integration at four under budget eight has no initial-area discrepancy because four is already an observed grid point.

Monotone nondecreasing utility implies that a left-step coarse area cannot exceed the complete state-path area. General repair does not guarantee monotonicity. Collateral degradation supplies the reverse-sign counterexample: a state with unaffected utility one can switch to zero between observations, making the coarse area too large. Nor does observing a successful final state rule out an unobserved transient failure or recovery. Full commit logs and fresh values are necessary for the stronger reconstruction claim.

Unknown state-changing events, an incomplete routing/environment snapshot, a fresh same-state response that differs from the earlier response, a missing positive-duration state audit, or a discrepancy exceeding the geometric bound falsifies the corresponding identification step. Such cases should suspend the point estimate; a critic's agreement cannot repair missing observations. Our self-checks include missing positive-width audits, missing terminal-only observations, a decreasing-utility counterexample and unchanged-state commits.

$$E = \left(\frac{1}{B}\right) \sum_{j=0}^{J-1} (g_{j+1} - t_j) \tag{38}$$

$$|\Delta_A| \le E \tag{39}$$

$$A_{\text{lower}} = A_{\text{observed}}, \quad A_{\text{upper}} = A_{\text{observed}} + \frac{W_{\text{unknown}}}{B} \tag{40}$$

24.4 Preserved v19: Sharp audit-only bounds with shared state

Fix one audit panel, budget B and the twelve equally weighted authored templates. For each policy p and complete state identity s, let w(p,s) be the normalized time spent in that state, averaged over templates. The panel utility u(s) is binary and shared across visits only under the complete-state, pure deterministic worker and fixed-input assumptions. A known state hash alone does not establish those assumptions. In this new analysis the source observations are checked for concordance, then deliberately hidden according to a declared view.

The information model knows commit times, state identities, policy occupation weights and revealed audit utilities. It does not use the arithmetic source program to infer a hidden score. This restricted audit-only model is essential: the researcher can read the complete fixture and the full saved outcomes, so the experiment measures robustness to controlled disclosure, not irreducible ignorance about an inaccessible deployed system.

Write the neutral-minus-selective mean-area contrast as a signed sum with coefficient c(s)=w(neutral,s)-w(selective,s). Revealed scores contribute K. Each unrevealed utility can be zero or one, subject to equality across the same state. A positive coefficient is minimized at zero and maximized at one; a negative coefficient has the opposite assignment. The resulting endpoints are jointly attainable within this audit-only model. With binary scores the set of compatible contrasts can be discrete: we claim a sharp interval hull, not that every interior real value is attainable.

Bounding each policy independently and subtracting its intervals forgets that the same unknown utility appears in both policies. The direct interval is always contained in that subtraction interval. Exact cancellation does not require that either policy area be identified. This is elementary finite linear optimization over a cube; no priority claim, causal effect identification or external theorem transfer is made.

$$A_p = \sum_s w_{p,s} u_s \tag{41}$$

$$D = \sum_s c_s u_s, \quad c_s = w_{n,s} - w_{q,s} \tag{42}$$

$$K = \sum_{s \in O} c_s u_s \tag{43}$$

$$\ell = K + \sum_{s \in M} \min(c_s, 0) \tag{44}$$

$$r = K + \sum_{s \in M} \max(c_s, 0) \tag{45}$$

25 Preserved v19: Experimental Setup

We distinguish the original policy experiment from a later measurement experiment on its frozen paths. Every number below comes from the two retained local execution result files, not an external paper or a new model call. The editorial v13 cycle changes the exposition, figure specification and reviewed references; it does not create new policy trials.

25.1 Preserved v19: Authored cohort, policies and budgets

The cohort comprises twelve authored templates: eight contain a previously rejected exact candidate and four contain no useful rejected candidate. Candidate position, hint alignment and presence of the written fixture lesson vary. Each archive has four candidates. The same six unique arithmetic inputs serve development, two additional-input checks, the initial audit, later-input audit and unaffected-input audit. Reuse of those inputs across policies does not create independent tasks or histories.

The old receipt labels one call kind independent_authored_qualification. That historical identifier is retained for byte compatibility, but its meaning is restricted here to two additional authored numeric inputs that the controller does not read. The same author, source code and arithmetic rule determine every input and answer. Those checks are neither independently authored correction qualification nor a secret held-out evaluation. Programmatic non-use by the controller is weaker than an access-controlled external audit boundary.

All seven policies receive work-unit caps of five and eight. Rescoring costs one authored unit, a fresh candidate execution two, integration one and lesson treatment one. Selection reserves enough allowance for its fresh execution/integration and, where assigned, lesson action. It stops at the first development-positive candidate, freshly executes that candidate and integrates on development success. Neutral and lineage-first policies share this acceptance rule. Equal caps do not force equal spending. No candidate is newly generated during these fixed-pool procedures.

The selection-by-lesson factorial uses no repair, selection only, lesson only and selective joint repair with V_new in every cell. This is a whole-procedure contrast: data-query ordering, failed selection, reservations and integration can differ across cells. Neutral early-stop and neutral joint are essential additional controls; exhaustive replay alone would be a weak comparator. The rescore-only arm is an endpoint control rather than a candidate recovery procedure.

Policy	Archive/selection action	Lesson action
No repair	None	Retain
Rescore only	Score immutable stored responses	Retain
Neutral early-stop	Fixed archive order, first development-positive candidate	Retain
Selection only	Hint-first order, same acceptance rule	Retain
Lesson only	Keep incumbent	Suspend if active
Neutral joint	Neutral early-stop with lesson reserve	Suspend if active
Selective joint	Hint-first selection with lesson reserve	Suspend if active

Table 13: Same starting history and allowed information; the realized costs and outputs remain outcomes.

25.2 Preserved v19: Frozen-trajectory committed-state re-audit

Before the follow-up, the runner verifies every artifact hash in the prior v11 execution receipt and pins the old worker implementation. It fixes all 168 trajectory files, twelve history records, the three audit inputs and its own analysis code. Every event is replayed from the recorded initial state. Integration checks the archive artifact and preserved valid change; lesson treatment checks its supporting text hash. Reconstructed pre-lesson and final snapshots must exactly equal the authenticated original snapshots. An unrecognized event is rejected because it could hide a state mutation.

For each trajectory separately, every distinct committed-state hash is submitted to a new local worker subprocess on the initial, later and unaffected inputs. There is no audit cache across trajectories. A repeated byte-identical state within a trajectory reuses its new observed result only under the declared deterministic-worker assumption; this reduces redundant subprocesses without pretending that reused observations are independent samples. Every request, stdout, stderr, worker hash, process return code and elapsed duration is preserved.

The new observations are compared with the original coarse observations wherever they bind the same state hash. Drift suspends commit-aware point estimation. The primary analysis compares A_C with the retained A_G on the same frozen policy path, checks the geometric bound for all three

panels and reports every initiated template. No policy sees the fresh audit outputs, no candidate is regenerated, no lesson treatment is changed and no failed policy is removed.

The first follow-up attempt stopped after one state subprocess because the adapter initially omitted the reused area function's explicit budget argument. That failed directory, its frozen plan, exact failed analysis-code bytes and failure record are preserved. The corrected complete attempt uses a new directory. Its 292 state subprocesses are reported separately from that one completed subprocess in an incomplete attempt. No incomplete attempt is counted as a successful trajectory or discarded from the execution history.

Verify prior receipt hashes and freeze all trajectory/history inputs before fresh execution.
Reconstruct every state commit and validate initial, pre-lesson and final checkpoint hashes.
Within each frozen trajectory, newly execute every distinct committed state on the same three panels.
Bind fresh outputs to request/state/worker hashes and check agreement with prior same-state observations.
Compute coarse and committed-state areas, missingness bounds and the geometry error bound.
Report all 168 paths, separate audit resources and retained failed attempts; do not alter the policies.

Algorithm 3: A post-freeze measurement study, not a second policy-performance trial.

25.3 Preserved v19: Analysis unit and resource accounting

The first experiment contains twelve authored history templates, seven procedures and two caps: $12 \times 7 \times 2 = 168$ trajectories. They reuse six unique authored arithmetic inputs. The state follow-up reuses three of those panel inputs, auditing 292 distinct states within trajectories and producing 876 outputs. Neither repeated policies, panel outputs nor subprocess invocations enlarge the independent history sample. The template construction varies availability of a useful rejection, lesson presence, archive order and aligned/misleading lineage; it does not sample those factors from deployment.

The action schedule charges one unit per rescore, two per candidate invocation, one per integration and one per lesson treatment. These are authored work units. We report realized spending under equal caps, because forcing equal spending would remove early stopping as an outcome. Common external audits have a separate ledger: the first experiment used 840 coarse-checkpoint calls plus 24 same-artifact memory audits; its total 1,048 subprocesses also includes candidate/qualification work. The successful follow-up uses 292 state subprocesses. No provider costs or hardware-calibrated efficiency ratios are inferred.

All selected tables are generated from frozen summaries, with all twelve started histories in each row. A zero conditional-recovery denominator is null. No-repair, rescore-only and lesson-only arms remain even when their recovery area is zero. The incomplete first follow-up attempt remains outside successful aggregates and is explicitly reported in the reproducibility appendix.

25.4 Preserved v19: Controlled omission protocol

Before calculating omission outcomes, a new plan and exact analysis/test code are frozen with byte snapshots of all 168 v12 traces, their receipts/results, the original worker and the v13 paper/source/claim records. The new run uses only the Python standard library and invokes neither a worker nor a model endpoint. All prior receipt-listed outputs are authenticated. Current source files are rechecked after the analysis. The successful exact computation takes about 12.89 seconds on one local host; this is not an algorithmic efficiency comparison.

There are nine unique complete state hashes across the entire fixed cohort. A revealed global state audit discloses its three fixed-panel scores and can inform every identical-state visit under the declared deterministic contract. We enumerate all 512 observed-state subsets for each budget and panel. This produces 3072 bound rows, not 3072 independent tests. The earlier v12 execution deliberately made no cross-trajectory audit cache; reusing its now-available deterministic observations is a new analysis rule, not a retrospective reduction of that execution cost.

Six predeclared views reveal all states, no states, initial states, terminal states, prior coarse-grid states, or states whose first hexadecimal hash digit is even. These rules inspect identity, position or time, not score values. They are controlled sensitivity interventions; no missing-at-random mechanism or probability of real omission is fitted. For each subset the formula is compared with exhaustive compatible binary completions, the full-data value must be contained, and revealing another audit must not widen the interval.

Retrospective certificate searches report the smallest revealed-state count that exactly identifies a contrast, establishes neutral noninferiority, establishes strict neutral advantage, or identifies both individual areas. A certificate uses the known source outcomes to evaluate candidate observation sets. It is not an adaptive sampling algorithm, a prospective stopping rule or a demonstrated way

to reduce real audit expenditure. Source outcomes never guide a new policy, and no frozen recovery decision is changed.

26 Preserved v19: Results and Analysis

We report the endpoint separation and policy negative result before the new area analysis. This ordering prevents a more informative measurement from being mistaken for an improved controller.

26.1 Preserved v19: Saved-score improvement is not current-state recovery

The completed v11 run contains 1,048 local subprocess calls and no model requests. It reports 168 policy-budget trajectories, not an empirical sample size of 168 independent histories. All twelve templates improve at least one saved score after correction, but the rescore-only procedure leaves all current states unrecovered. Fresh development executions also expose endpoint non-equivalence: twelve successful candidate invocations in each unrestricted selection arm yield only four initial-input successes. The fixture's memorizing candidate explains that constructed generalization failure.

At budget eight, neutral joint and selective joint both have four initial successes and four later-input successes out of twelve. Without lesson suspension the corresponding selection procedures have four initial successes and two later-input successes, giving a conditional maintenance-failure value of 2/4. Procedures with no initial successes have null conditional maintenance failure. These results do not demonstrate a general advantage of lineage-first repair.

B=8 procedure	Saved-score improvement	Fresh dev candidate success	Initial state success	Later-input state success
no repair	0/12	No call	0/12	0/12
rescore only	12/12	No call	0/12	0/12
neutral early stop	12/12	12/12	4/12	2/12
selection only	12/12	12/12	4/12	2/12
lesson only	0/12	No call	0/12	0/12
neutral joint	12/12	12/12	4/12	4/12
selective joint	12/12	12/12	4/12	4/12

Table 14: Version 11 authored executor results. Different columns are different endpoints; none is a language-model accuracy estimate.

26.2 Preserved v19: A strong neutral baseline limits the policy claim

The terminal-count tie does not justify declaring all procedures equivalent: realized cost and collateral capability remain outcomes. Table 3 shows the joint procedures. Under cap five, neutral joint has the same four initial and four later-input successes, spends three rather than five authored units on average, and preserves the unaffected task in all twelve histories rather than four. Under cap eight, terminal counts and collateral loss tie, with lower authored spending for selective joint. This reversal is why we do not claim general lineage-first superiority.

The selected procedure follows an authored hint. In the six aligned-hint histories its later-input success is 4/6 versus 2/6 for neutral joint; in the six misleading-hint histories it is 0/6 versus 2/6. Aggregate equality is partly a consequence of the deliberately balanced construction. The same-artifact lesson fork provides a narrower mechanism contrast: two of twelve retain/suspend pairs improve on the later input and ten do not. The programmed lesson rule explains this result; it is not a discovered causal property of real model memory.

Cap	Procedure	Initial / later successes	Mean work units	Unaffected loss
5	neutral joint	4/12 ; 4/12	3.00	0.000
5	selective joint	4/12 ; 4/12	5.00	0.667
8	neutral joint	4/12 ; 4/12	5.67	0.667
8	selective joint	4/12 ; 4/12	5.00	0.667

Table 15: Executed authored paths, twelve histories per row. Work units are not tokens, money or CPU. The budget-five neutral advantage is retained.

26.3 Preserved v19: Fresh state audits reveal two directions of sampling error

The successful follow-up reconstructs all 168 trajectories and makes 292 fresh state subprocess calls, producing 876 panel outputs with no failed calls. Every comparable same-state output agrees with the prior observation. The earlier coarse protocol made 840 checkpoint audit calls on these trajectories; its additional 24 fixed-artifact memory-audit calls are excluded from that schedule comparison. The 292 count exploits deterministic within-trajectory state identity, so it is not a universal audit-cost advantage for stochastic agents. Observed subprocess duration sums to approximately 57.30 seconds in this execution, without a controlled latency comparison.

Sixteen trajectories have positive initial-area discrepancies, each exactly 0.2; the other 152 have zero difference. All positive cases are budget-five procedures with successful integration at cost four. Per procedure, four of twelve templates change from coarse initial mean zero to committed-state mean 1/15. At budget eight the corresponding initial mean remains 1/6 because integration is grid-aligned. Each positive initial discrepancy attains its trajectory's geometry bound. Terminal success counts remain exactly those of v11.

Later-input utility exposes a separate timing issue. At budget five, joint repair can suspend a harmful lesson only at the cap five. That increases the final later-input success count from two to four without adding positive-duration utility after that last treatment. The committed-state later-input mean is consequently 1/30 for both single-selection and joint procedures at that budget. At budget eight, joint repair's later-input area changes from 1/8 under coarse observation to 7/48 under the full commit path, because lesson suspension occurs at five between the coarse points four and six.

Unaffected-task utility has negative discrepancies in 32 trajectories and zero in 136. For example, at budget five the selective-joint mean drops from coarse unaffected utility 1 to committed-state utility 13/15, while neutral joint remains at 1. At budget eight, neutral joint changes from 5/6 to 3/4 on that panel. All panel discrepancies satisfy their geometry bounds. These reverse-sign cases refute a universal claim that coarse left-step evaluation is conservative. The policy ties, misleading-lineage failures and budget-five neutral-baseline advantage are preserved rather than reinterpreted as a new repair improvement.

Budget/procedure	Initial A_G	Initial A_C	Later A_G	Later A_C	Fresh audits / prior coarse
5 / neutral early-stop	0	1/15	0	1/30	24 / 60
5 / selection only	0	1/15	0	1/30	24 / 60
5 / neutral joint	0	1/15	0	1/30	22 / 60
5 / selective joint	0	1/15	0	1/30	30 / 60
8 / neutral early-stop	1/6	1/6	1/12	1/12	24 / 60
8 / selection only	1/6	1/6	1/12	1/12	24 / 60
8 / neutral joint	1/6	1/6	1/8	7/48	30 / 60
8 / selective joint	1/6	1/6	1/8	7/48	30 / 60

Table 16: New observations of the frozen authored paths, twelve templates per row. No-repair, rescore-only and lesson-only initial/later areas remain zero at both budgets. Their audit counts are retained in results.json, not dropped.

26.4 Preserved v19: Reading the area and terminal outcomes together

Figure 2 isolates the cost-four integration under cap five. The last coarse left endpoint is 3.75, so a successful [4,5) interval contributes nothing to the coarse statistic and one fifth to the complete committed-state area. Across the frozen cohort, this occurs on sixteen initial-utility paths. Figure 3 contrasts initial and unaffected panels for neutral and selective joint repair; the latter can also deteriorate between observations. Terminal counts do not change when the audit schedule is refined.

The new calls agree with the earlier responses whenever they bind identical state bytes. Every observed panel-area discrepancy lies within the geometry bound. These checks support the internal reconstruction claim for this worker and panel; they do not certify that three inputs characterize a general program or a language model. A final successful state can still have zero post-treatment area if treatment completes exactly at the cap, and an apparently stable coarse curve can still hide a collateral failure.

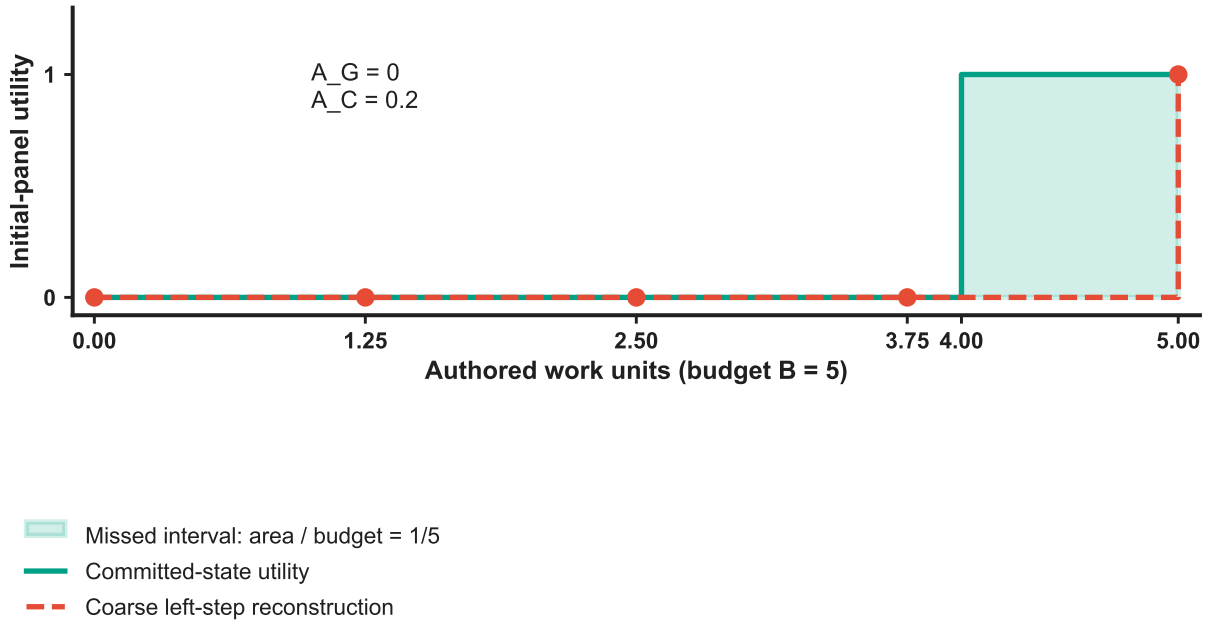


Figure 3: A constructed cost-indexed state path, verified by fresh state execution. Integration at cost four under cap five yields committed-state area 0.2 but coarse left-step area zero. The shaded missed interval attains the geometric bound. Assumptions are complete commits, pure deterministic execution and a fresh observation of every positive-width state.

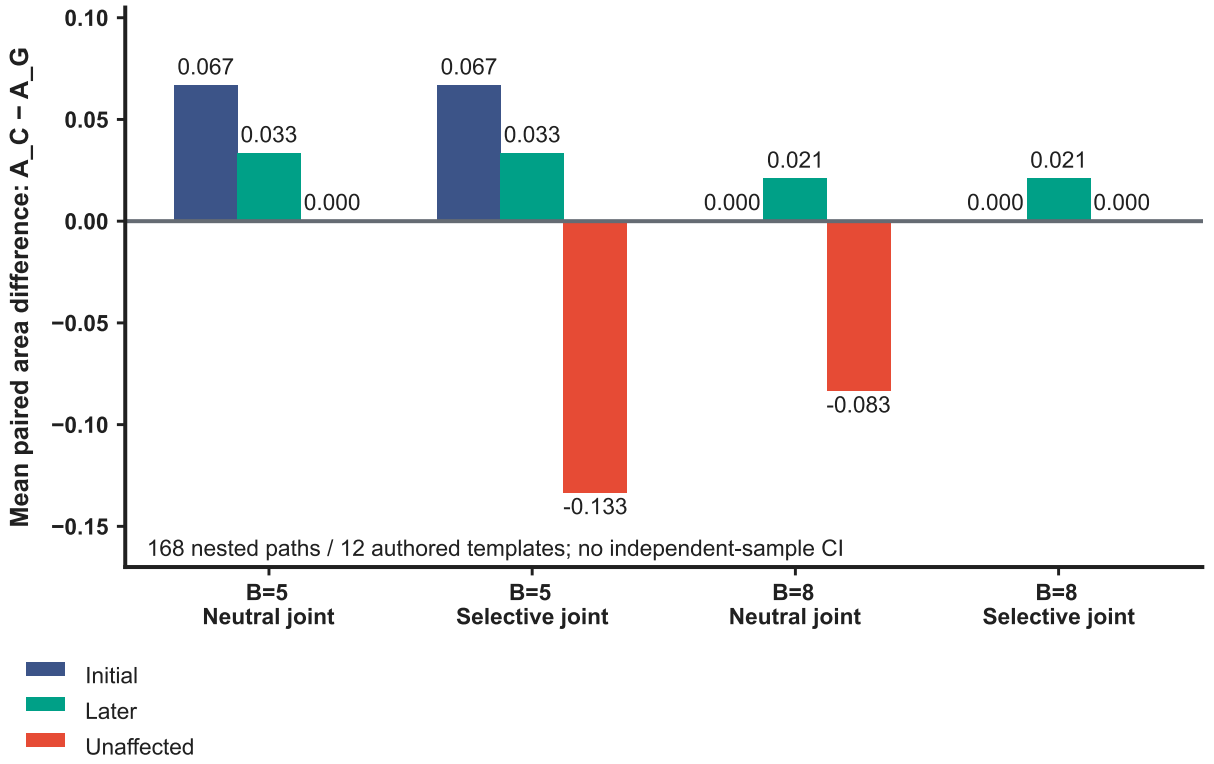


Figure 4: Paired coarse and committed-state mean utility on fixed authored paths, twelve history templates per procedure/cap. Initial utility can be underestimated; unaffected utility can be overestimated. Across all 168 nested paths, sixteen initial areas increase and thirty-two unaffected areas decrease. These are descriptive finite-cohort counts, not independent population samples or LLM effects.

26.5 Preserved v19: A policy contrast can outlive missing individual areas

Figure 4 compares bounds for the budget-eight unaffected panel. With no revealed audit, the shared-state contrast lies between -1/12 and 1/12, substantially narrower than the naive subtraction range from -1 to 1, but its sign remains ambiguous. Revealing the two initial states yields a neutral-minus-selective bound from 0 to 1/12. The predeclared six-state hash-even view strengthens the lower

endpoint to 1/24, certifying strict neutral advantage within the audit-only model. Naive independent subtraction still includes negative values, from -7/48 to 13/48.

At budget five the same six-state view identifies the initial and later-input contrasts exactly as zero, even though separate-area subtraction spans from -1/15 to 1/15. It identifies the unaffected contrast as 2/15. Under full observation the budget-eight initial/later contrasts are zero and the unaffected contrast is 1/12. Thus better handling of dependence preserves both exact ties and a neutral advantage; it does not create a win for selective repair.

All terminal-only and prior-coarse global views happen to reveal every one of the nine states somewhere in the cohort. Under known complete commit times they therefore identify the same exact areas as the all-observed view. This does not vindicate a left-held coarse interpolation or identify missing commit times. It shows why omission must be defined globally: treating a state as unknown in one path while revealing its identical deterministic outcome elsewhere would manufacture uncertainty.

B8 unaffected audit view	Revealed states	Direct sharp endpoints	Separate-area subtraction
All	9	1/12 ; 1/12	1/12 ; 1/12
None	0	-1/12 ; 1/12	-1 ; 1
Initial only	2	0 ; 1/12	-5/12 ; 1/2
Hash-even	6	1/24 ; 1/12	-7/48 ; 13/48

Table 17: Normalized neutral-minus-selective utility-area endpoints on the same twelve authored templates. These are identification bounds under an audit-only model, not confidence intervals.

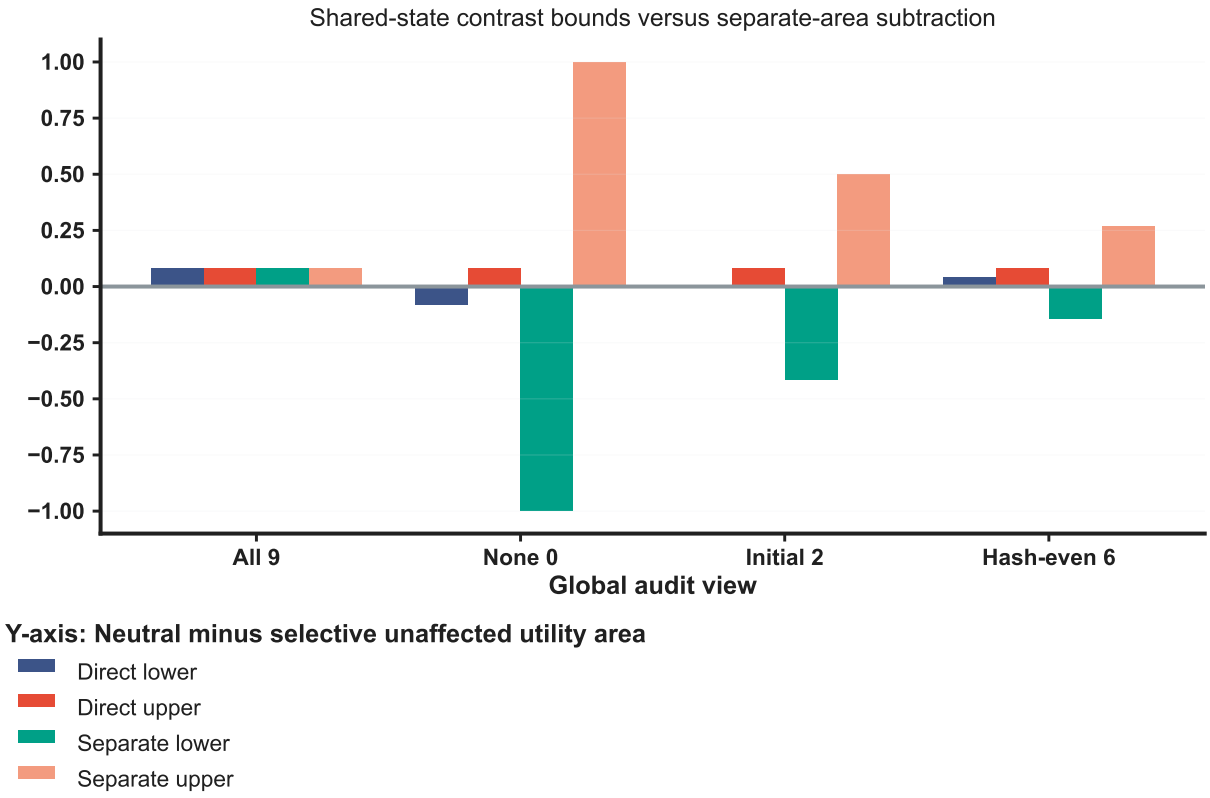


Figure 5: Budget-eight unaffected-area contrast under four predeclared global audit views. The direct shared-state endpoints preserve dependence; separate-area subtraction discards it. The bar values are interval endpoints, not independent outcomes. Complete commit times and deterministic state equality remain assumptions.

26.6 Preserved v19: Exhaustive audit subsets and retrospective certificates

Figure 5 summarizes all observation subsets by their size. Across all 512 sets, budget-five initial and later contrasts are exact ties in 16 sets and sign-ambiguous in 336; the unaffected contrast is strictly neutral-positive in 128 sets and sign-ambiguous in 288. At budget eight the corresponding initial/later exact-tie counts are 32 and ambiguous counts 288; unaffected strict-positive and ambiguous counts are 160 each. The remaining sets support one-sided nonstrict statements. These finite counts do not assign a distribution to missingness or imply a success probability.

The smallest number of revealed state audits for exact neutral-minus-selective contrast is five at budget five and four at budget eight, for every panel. Identifying both separate policy areas needs seven and eight states respectively. Strict neutral advantage on the unaffected panel can be certified

with three revealed state audits at either budget. No audit subset can certify strict neutral superiority on the initial or later panel, because the true complete-data contrast is zero.

Among the 512 subsets for each panel, the direct interval is strictly narrower than naive subtraction for 480 sets at budget five and 510 at budget eight. Twelve sets at budget five and thirty at budget eight identify the contrast while leaving at least one individual area unidentified. Every one of 3072 bounds equals exhaustive binary-completion extrema;6144 endpoint assignments attain the claimed endpoints, and 13824 one-state reveal edges satisfy monotone narrowing. These are finite contract checks, not independent scientific replications.

Budget/panel	Exact contrast: minimum audits	Both areas: minimum audits	Strict neutral advantage: minimum audits
5 / initial or later	5	7	Not possible
5 / unaffected	5	7	3
8 / initial or later	4	8	Not possible
8 / unaffected	4	8	3

Table 18: Retrospective minima over the nine global complete-state audits and all512 subsets. One state audit reveals three fixed-panel scores. The v14 subset analysis did not execute a scheduler; the v15 known-path replay is reported separately below.

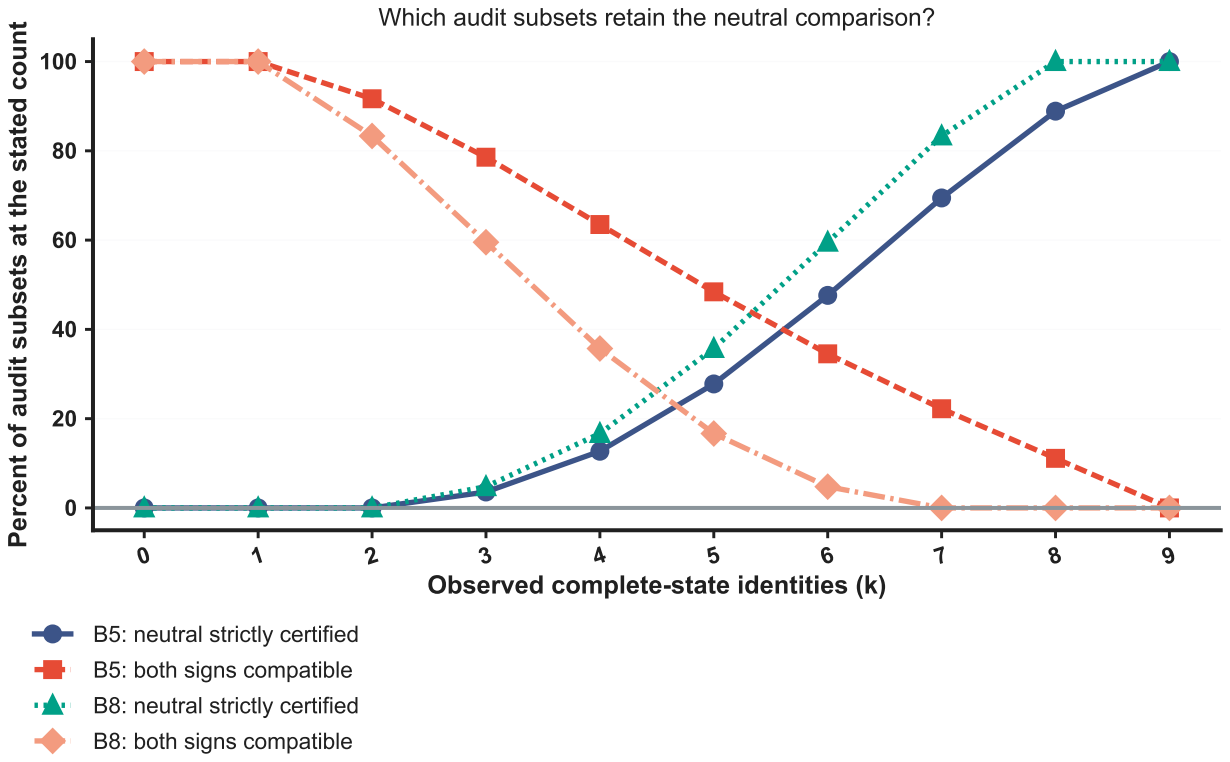


Figure 6: Unaffected-panel audit-subset classification by the number k of revealed states. At each k the denominator is choose(9,k); plotted fractions are exhaustive combinatorial summaries, not empirical omission probabilities or independent model tests.

27 Preserved v19: utility-blind ordering on complete known paths

For a fixed budget and neutral-minus-selective contrast, each complete state has a signed occupation coefficient c_s. Under the same binary audit-only information model, an unrevealed state contributes interval width |c_s|. Revealing one state eliminates exactly that term and discloses all three fixed-panel utilities. Therefore selecting the q largest absolute coefficients minimizes remaining width among all q-state audit sets. The result concerns width, not expected predictive value or a learned audit policy.

The proof is a finite exchange argument. If a selected state has smaller absolute coefficient than an unselected state, swapping them cannot increase the removed width. Repeating such exchanges gives decreasing absolute coefficient order. Ties use ascending SHA-256. Zero coefficients are queried last. At q equal to the number of nonzero coefficients the direct contrast is exact, even when separately identified policy areas still require more states.

We compare three utility-blind schedules: target_abs_coefficient, separately for each budget; cross_budget_l1_coefficient, which sorts by the sum of absolute coefficients over budgets five and eight; and state_hash. A fixed complete-state audit reveals initial, later and unaffected panel values together at one quote unit. Because coefficients depend on path occupation and policy weights rather than the panel score, no panel utility is needed to form an order.

The global cross-budget score is a sharing compromise, not a guarantee of optimality for each individual budget. Audit outputs are reused by complete state identity under a deterministic worker and fixed panel inputs. We do not charge three queries for three panel values, and do not count a state occurring in many of the 168 paths as many distinct new audits. Query quote units are an explicit synthetic cost model; no actual worker or audit execution is made in this replay.

Input bytes, plan and exact code are frozen in a new directory before the query sequence is analyzed. We reuse all 168 earlier complete paths from 12 authored histories and nine unique states. The coefficients include equal weights over all twelve templates and preserve every path. Six target contracts comprise two budgets times three panels, and each schedule is evaluated at all ten prefixes q=0 through9. The resulting 180 rows are dependent audit diagnostics, not 180 policy trials.

$$W_b(O) = \sum_{s \notin O} \left| c_{b,s} \right| \tag{46}$$

$$s_{q+1} = \arg\max_{s \notin O_q} \left| c_{b,s} \right| \tag{47}$$

$$q_{\text{exact},b} = \left| \left\{ s : c_{b,s} \neq 0 \right\} \right| \tag{48}$$

27.1 Preserved v19: Width optimality does not imply minimum sign-certificate cost

We exhaust every q-state subset for both target budgets and verify the greedy prefix width against its exact minimum in twenty cardinality checks. All 180 replay intervals contain the complete-data contrast and shrink monotonically as audits are revealed. A second implementation reconstructs occupation coefficients from the raw paths and checks every bound against 18,414 compatible binary completions. These are alternative computations on the same authored data, not independent scientific replication.

Target coefficient order identifies the exact contrast after five state queries at budget five and four at budget eight. Hash order requires seven and eight respectively. The cross-budget shared order needs five and six. The latter's extra budget-eight queries expose a real multi-target allocation tension: one order serving two contrasts can be less efficient for one target than its individually optimal order. These are exact quote counts in the fixed known-path model; they do not measure physical or model-inference savings.

A stopping rule may inspect revealed bounds and stop when the lower endpoint is strictly positive, the upper endpoint strictly negative, or the interval is a point. It cannot inspect unrevealed utilities. The initial and later true contrasts are zero, so all schedules must reach a point to certify that exact tie. The unaffected panel has positive truth; target coefficient order certifies strict neutral advantage at q=4 for budget five and q=3 for budget eight. Hash order takes q=5 and q=7, and the cross-budget order q=4 for both.

Exhaustive hindsight subset search needs only three audits to certify positive unaffected contrast at either budget. Thus the budget-five width-optimal schedule takes one extra query relative to the outcome-informed optimum. This does not contradict width optimality: sign certification depends on the revealed values and the direction of coefficients, while width depends only on coefficient magnitudes. Retrospective minima cannot be reported as a utility-blind scheduler's realized cost.

The observed true contrast remains the original authored-fixture outcome. No neutral policy receives a new success credit because the audit schedule is efficient. Likewise a narrower interval does not improve the underlying recovery worker. This continuation changes which already saved state audits need to be read to support a statement, not what policy did to the current implementation or lesson memory.

Budget / schedule	Exact contrast queries	Unaffected strict sign queries	Hindsight sign minimum
5 / target_abs_coefficient	5	4	3
5 / cross_budget_l1_coefficient	5	4	3
5 / state_hash	7	5	3
8 / target_abs_coefficient	4	3	3

Budget / schedule	Exact contrast queries	Unaffected strict sign queries	Hindsight sign minimum
8 / cross_budget_l1_coefficient	6	4	3
8 / state_hash	8	7	3

Table 19: Saved-audit replay on known complete paths; one state query reveals all three panel utilities. Hindsight uses full outcomes and is not a competing deployable schedule.

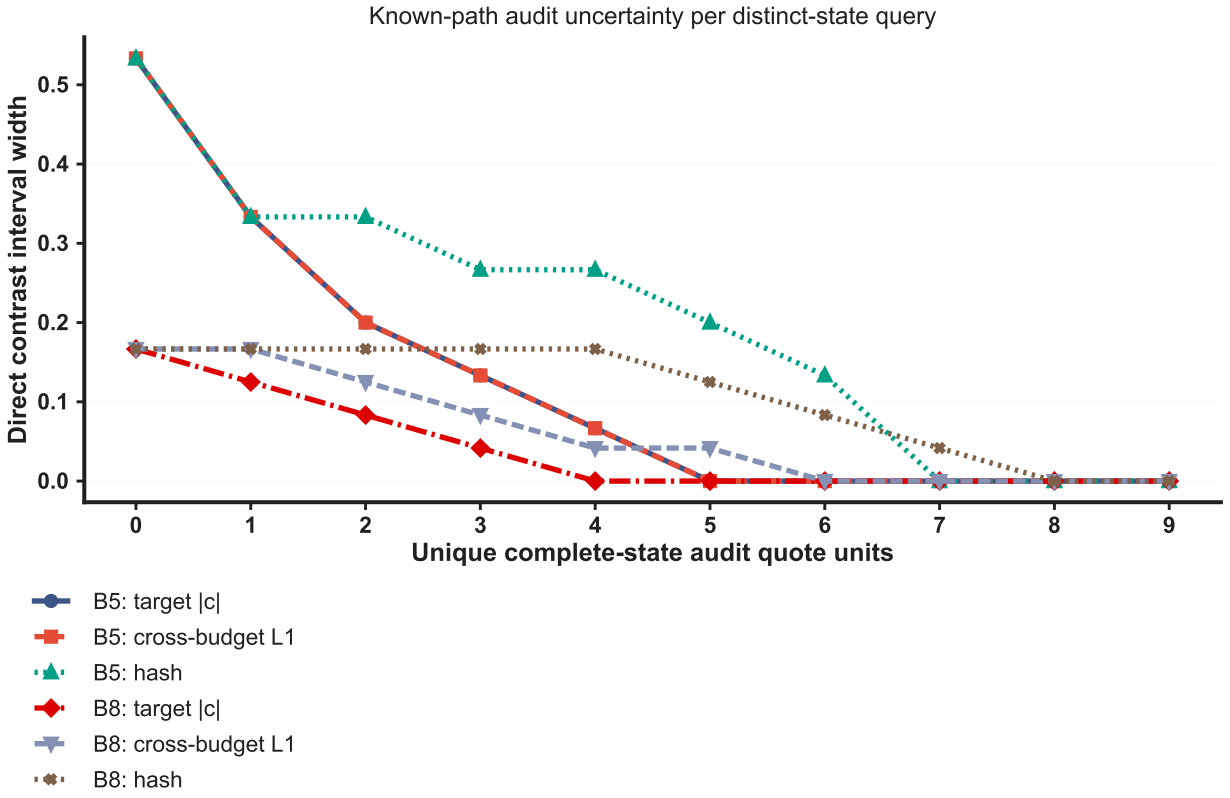


Figure 7: V15 known-path, utility-blind audit orders. The target-|c| rule attains the minimum possible direct-contrast interval width at every query count in its budget. Hash and cross-budget orders are controls. All costs are distinct-state query quotes from cached audit replay; no new audits, unknown-future-path trial or latency measurements.

27.2 Preserved v19: Information available before a query is the decisive boundary

The word prospective must be scoped carefully. The order is fixed before disclosing the state utilities in this replay, but the complete trajectories and commit times already exist and their earlier outcomes have been exposed to the researchers. The schedule is implementable for a post-run auditing task with fixed known paths. It is not presently a scheduler that chooses audits while recovery is still making unknown future commits.

For an online recovery auditor, a new state changes the future occupation coefficients, and its duration is unknown until the next commit or terminal budget. An order computed from complete future paths would leak information unavailable at that decision time. A faithful online extension would need a predictable allocation rule based on current logs, an explicit update of coefficient uncertainty and a stopping definition valid under changing path geometry. That extension was not implemented or claimed here.

All direct-bound guarantees also require complete commit logs, complete state identity, deterministic fixed-input utilities and no hidden mutation between logged states. Hash equality alone does not prove semantic equivalence or determinism. If hidden commits or stochastic audit drift are possible, the v14 failure counterexamples remain active and a narrow shared-state interval can be invalid. A scheduler must not conceal a failed identification assumption by continuing to report precise bounds.

A genuine resource evaluation should log model/worker invocations, elapsed time, cache lookups and failed queries in addition to unique-state quote units, and preserve natural histories with evaluator corrections rather than constructing only known arithmetic defects. Those records remain absent. The next admissible step is a fully instrumented natural-history run with complete current-state audits and a known-path audit order fixed before audit outcomes are revealed; long-term recurrence and online scheduling require additional independent designs.

28 Preserved v19: auditing without future commits

A deployable audit decision cannot depend on a commit that has not arrived. V15's known-path coefficient order remains a valid offline comparison, but it cannot be used unchanged when the final occupation coefficients are unknown. V16 therefore executes three explicitly causal decision rules through a prefix-only interface, while retaining the earlier signed-state target and every authored-template denominator. The change concerns audit information allocation; it does not create another recovery worker or new failure histories.

For each recovery horizon B in {5,8}, the evaluator reveals all commits through integer checkpoint t, then permits at most one distinct-state quote. The total quote cap is q in {1,2,3,4}. A quote returns the three cached fixed-panel utilities for one observed complete state. The recovery stream is fixed and does not wait for the audit. One quote per checkpoint is a stipulated opportunity constraint, not measured asynchronous latency; these extra audit quote units are not subtracted from the original recovery cost horizon. Identical arrivals, opportunities and caps apply to every competing rule.

The policy sees only timestamps and SHA-256 state identities already visible through t, the completed-prefix signed occupation coefficients, the currently active arm-state frequency difference, and its previously quoted identities. Future commits, terminal coefficients, unrevealed utility values and state program semantics are absent from the decision interface. Utilities enter only the subsequent bound computation after a query has been selected. The evaluator may inspect the saved full trace to score a decision, but that capability is not passed to the policy.

The prefix occupation coefficient d_s(t) averages over all twelve templates in each arm and integrates only completed recovery cost [0,t). Its normalization remains the full fixed horizon B, rather than renormalizing the partial trajectory to one. Thus prefix utility and the remaining future contribution add to the original final contrast. The currently active frequency coefficient a_s(t) describes only the arm difference at t; using it to rank a query does not assert that the state will persist.

The first_seen baseline eagerly queries the earliest arriving unaudited identity, including identities whose present coefficient cancels. The prefix_abs rule queries the largest nonzero |d_s(t)|, while current_disagreement queries the largest nonzero |a_s(t)|; each skips if no eligible identity exists and breaks ties by ascending SHA-256. These simple rules require neither a utility prior nor a learned model. They are predeclared heuristics under changing future weights, rather than optimal adaptive sampling policies.

$$d_s(t) = \left(\frac{1}{12B}\right) \sum_{h=1}^{12} \int_0^t \left(\mathbf{1}\left[S_{h,N}(z) = s\right] - \mathbf{1}\left[S_{h,L}(z) = s\right] \right) \mathrm{d}z \tag{49}$$

$$r_t = \frac{B - t}{B} \tag{50}$$

28.1 Preserved v19: Unknown-suffix bounds and an unavoidable sign ambiguity

Even exhaustive audits of already observed states do not reveal an unknown future path. At checkpoint t, we bound each policy's remaining normalized utility contribution between zero and r_t=(B-t)/B. Their difference is therefore in [-r_t,r_t]. Adding this suffix to the shared-state prefix interval gives a valid final-contrast interval without assuming that future states belong to the nine identities eventually present in the saved files. This is an outer bound for the online information model; we claim sharpness only for the terminal fixed-path audit interval when r_t=0.

For a quoted-state set O_t, let K_t be the signed prefix sum over revealed utilities. The lower endpoint adds negative unknown prefix coefficients to K_t and subtracts r_t; the upper endpoint adds positive unknown prefix coefficients and r_t. The same state/input deterministic equivalence assumption enables a quoted value to be reused when that state reappears. Hidden state mutation, incomplete commit logs or changing fixed-panel outputs invalidate this reuse and remain unresolved empirical admission conditions.

An explicit two-world construction illustrates why the future term cannot simply be deleted. Let B=4 and observe checkpoint t=1. Both arms have occupied the same audited utility-1/2 state, and no further commit has yet arrived. In one compatible completion, at cost 2 the neutral arm moves to utility 1 and the selective arm to utility 0; in the other, their future utilities are exchanged. The terminal contrasts are +1/2 and -1/2. Both completions agree with every observation through t=1. The stipulated prefix interval is [-3/4,3/4], containing both outcomes; its width is not claimed to be the tightest interval for a more restricted transition system.

Consequently, no rule using only that common prefix can universally certify the final sign. This is a constructive information limit, not a claim that every online instance is unidentifiable. A restricted

future transition graph, minimum dwell times, or a validated stochastic transition model could tighten the suffix. None is silently inferred from the exposed nine-state fixture. The witness is a separate analytic construction and is not counted as two additional empirical histories.

$$K_t = \sum_{s \in O_t} d_s(t) u_s \tag{51}$$

$$L_t = K_t + \sum_{s \notin O_t} \min(d_s(t), 0) - r_t \tag{52}$$

$$U_t = K_t + \sum_{s \notin O_t} \max(d_s(t), 0) + r_t \tag{53}$$

28.2 Preserved v19: Causal replay results reverse the greedy ranking

The frozen analysis reads all 168 old paths from twelve authored deterministic templates. Its neutral-versus-selective contrast uses 24 path records per horizon, or 48 across the two horizons; these records still refer to the same twelve templates. Three causal policies, four quote caps and two horizons yield 24 dependent replay configurations, 180 checkpoint decisions and 540 panel-bound rows. A separately written verifier integrates unit recovery-cost intervals, recomputes all bounds and decisions, and exhausts 744 candidate subsets across eight cap/horizon oracle contracts. Every bound contains the complete-data contrast; interval widths do not increase.

The clairvoyant comparator is computed by dynamic programming over queried-state subsets and checkpoint arrivals. It is allowed to skip a checkpoint and can use the final coefficient to choose which visible state to quote, but it cannot quote a state before arrival or use more than one slot per checkpoint. Its final-width minimum is independently checked by enumerating subsets and matching each sorted first-arrival time to a distinct available checkpoint. This lower bound isolates the information disadvantage; it is not a deployable policy, not a Bayesian optimum and not an outcome-specific minimum sign certificate.

At B=5 and q=4, current_disagreement reaches final width 1/15, equal to the arrival-constrained oracle. FIFO and prefix_abs both end at width 2/15. Prefix_abs uses only three of the four available quote units because the remaining visible prefix coefficients cancel when earlier slots are offered. All three rules obey the same cap; a cap is an upper bound rather than a command to spend a useless query. The unused credit and wider endpoint are both retained, rather than imputing an unexecuted fourth query.

The ordering reverses at B=8. With one quote, FIFO attains width 1/8 whereas prefix_abs and current_disagreement reach 1/6, a 1/24 disadvantage in width. With four quotes, prefix_abs reaches 1/24, FIFO 1/12 and current_disagreement 1/8; the oracle reaches zero. Thus the largest-past-coefficient rule can be worse than FIFO at a small cap and better at a larger cap on the same horizon. V15's exchange proof assumes fixed final coefficients and supplies no optimality guarantee for these evolving coefficients.

A narrower terminal width is not automatically an earlier sign certificate. Of the 24 replay configurations, two certify the positive unaffected contrast: current_disagreement at B=5/q=4, checkpoint 5, and prefix_abs at B=8/q=4, checkpoint 8. Both use four quotes. The initial and later contrasts remain zero, and none of the causal rules reaches an exact tie certificate within these caps. Every unreached target keeps a null first-checkpoint and null query count with the spent cap/quotes retained. These counts are inspection of cached utilities, not measured audit executions or natural recovery successes.

B / cap	FIFO width	Prefix width (used)	Current width (used)	Oracle width
5 / 1	7/15	1/3 (used=1)	1/3 (used=1)	1/3
5 / 2	1/3	1/5 (used=2)	1/5 (used=2)	1/5
5 / 3	2/15	2/15 (used=3)	2/15 (used=3)	2/15
5 / 4	2/15	2/15 (used=3)	1/15 (used=4)	1/15
8 / 1	1/8	1/6 (used=1)	1/6 (used=1)	1/8
8 / 2	1/12	1/8 (used=2)	1/8 (used=2)	1/12
8 / 3	1/12	1/12 (used=3)	1/8 (used=2)	1/24
8 / 4	1/12	1/24 (used=4)	1/8 (used=2)	0

Table 20: V16 final direct-contrast interval widths under identical visible-state arrivals and at most one quote per checkpoint. FIFO uses the full cap in all rows. Parentheses retain actual queries. Oracle knows final coefficients and is a lower bound; all other policies use prefix information only.

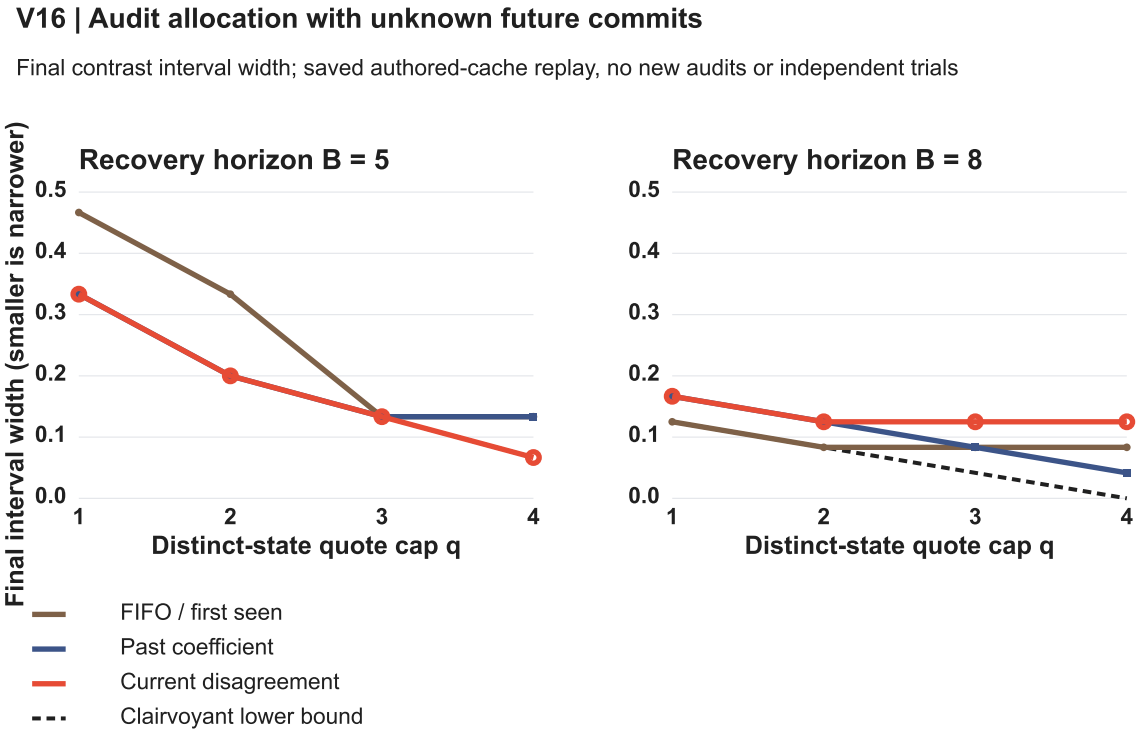


Figure 8: V16 causal audit replay: final interval width versus quote cap. The two horizons have different policy rankings; fixed-path greedy optimality does not transfer. Each point reuses the same authored cache. The dashed curve is an arrival-constrained clairvoyant lower bound, not a deployable comparator or a new experimental sample.

28.3 Preserved v19: Method comparison and a prospective admission test

Golovin and Krause's adaptive-submodularity framework [21] makes the policy depend on observed selected-item outcomes under a prior over realizations. We adopt the explicit partial-information boundary, but implement no Bayesian posterior or adaptive-submodularity approximation algorithm. Our final weights are revealed by future commits, and neither diminishing returns under a suitable realization model nor its required conditions is established. The author v5 revision also explicitly weakens a coverage guarantee after a proof flaw. We therefore do not describe our prefix heuristic as inheriting a general greedy guarantee.

The official COCO performance-assessment exposition [16] distinguishes fixed-budget indicator values from the evaluations needed to attain a target, retaining undefined runtimes for targets that are not reached. We similarly report final width at a quote cap separately from first sign/point checkpoint and quote count, but run no COCO functions, optimizer or restart protocol. In particular, checkpoint cost is recovery-stream geometry while a quote is a cached-state inspection; neither is presented as measured CPU/model latency or an externally comparable objective-call speedup.

The executable next experiment must freeze an audit selector before a genuinely unobserved commit stream begins, record each information view and decision, and log actual state execution calls, cache hits, failures and elapsed audit time separately. Recovery workers and audit allocations require different denominators. Replay feasibility is now checked; prospective operational cost and population recovery effects remain unmeasured. A claim of online savings is stopped if future commits or terminal coefficients enter the selector, or if repeated full-state/input audits disagree.

29 Preserved v19: robust allocation needs an explicit continuation class

The v16 ranking reversal leaves a methodological question: can a robust rule protect against an unknown continuation, or does it only optimize a convenient forecast? We retain the full 168 frozen paths and the same neutral-minus-selective contrast over twelve equal-weight authored template pairs at each horizon. No correction worker changes. The public audit contract is also unchanged: one previously unqueried visible state per integer checkpoint, no t=0 query, and at most q distinct-state quotes for q=1, 2, 3, 4. A quote reveals old saved fixed-panel values, not a new audit execution.

A selector receives only the horizon B, checkpoint t, commits visible through t, normalized prefix coefficients p, current arm-frequency difference d, first-arrival times and its own queried state identities. Future commits, terminal coefficients, source-program contents and utility payloads are absent. The remaining length is r=(B-t)/B. Unknown outcomes are disclosed to the interval evaluator only after a selection. Decisions are invariant when hidden future events and utility payloads are replaced while this visible view remains fixed.

First consider an unrestricted continuation that can transition arbitrarily soon to fresh state identities. The supremum worst final conditional audit-only width after querying set A is the unqueried prefix L1 mass plus 2 r. The upper bound is the triangle inequality. For the converse, future arm occupations can approach disjoint fresh states not in A; their opposing masses contribute 2 r without canceling any past state. This is a supremum or closure argument when the view already includes commits at t. We do not claim the supremum is attained under a positive minimum future commit delay. A single currently available query removes exactly its absolute prefix coefficient, so unrestricted one-step minimax is precisely prefix_abs, rather than a new universally stronger selector.

To obtain a different comparator we declare a narrower diagnostic class. Its final coefficient vector is c(lambda)=p+lambda*r*d with a common lambda from zero to one. Current arm occupations persist for that fraction of the remaining horizon and then their contrast cancels. This is a structured hypothetical continuation, not an inferred probability distribution or a claim that future states are confined to the nine eventually observed identities. The class can be wrong; its coverage of the saved terminal vector is tested separately.

For each final coefficient scenario, the residual binary-utility audit interval has width equal to the unqueried L1 mass. We minimize the worst such conditional width over the restricted segment. Convexity places its maximum at lambda zero or one, so exact evaluation requires the larger of two endpoint widths. This is not the width of the union of intervals when lambda itself remains unknown: known-state contributions can also shift with lambda. We use the restricted quantity to choose a query, never to certify a final sign or replace the unrestricted prefix-plus-suffix interval.

restricted_minimax_one selects the available state with the greatest strict immediate reduction in that endpoint maximum, using ascending SHA for ties; if every gain is zero it waits. restricted_minimax_set enumerates subsets of currently visible available states up to the remaining quote cap and public checkpoint opportunities, minimizes the endpoint maximum, prefers fewer quotes and a lexicographically earlier subset, and queries the first member now. It replans at the next checkpoint. The finite subset optimum is exact for that visible static decision, not a globally optimal online policy.

We retain FIFO, prefix_abs and current_disagreement unchanged, add persistence_abs ranking by absolute p+r*d, and add late_prefix. The latter waits until the last q scheduled checkpoints and then applies prefix_abs. It knows the horizon and public opportunity calendar but no future commit contents. All seven selectors share the same arrival rights, q cap and single-query opportunities; doing fewer queries is allowed and charged explicitly. Delayed selectors must pay in certificate timing and intermediate uncertainty, not just a final-width table.

$$c(\lambda) = p + \lambda r d \tag{54}$$

$$W_{\text{free}}(A) = \sum_{s \in A^c} |p_s| + 2r \tag{55}$$

$$W_{\text{restricted}}(A) = \max\left(\sum_{s \in A^c} |p_s|, \sum_{s \in A^c} |p_s + r d_s| \right) \tag{56}$$

29.1 Preserved v19: Model misspecification and a complementarity counterexample

The restricted segment fails to contain the actual final coefficient vector in nine of thirteen nonterminal views, including both initial t=0 views. Restricting the denominator to preterminal query checkpoints gives seven violations among eleven views. These views are dependent snapshots of two already exposed aggregate paths, not independent model failures or an estimated misspecification probability. The membership test uses final coefficients only in retrospective evaluation; it is never an input to a selector. Every reported sign certificate and truth-containing bound continues to use the unrestricted future suffix.

Membership is checked coordinate by coordinate over the union of prefix, current and terminal state identities. For a nonzero slope, every coordinate requires the same lambda; zero-slope coordinates must already equal the final coefficient. We also minimize L1 distance to the segment over its exact

piecewise-linear knots. At B=5, views t=0 through 3 are outside; at B=8, views t=0 through 4 are outside. The terminal views are trivially inside because r=0 and the prefix is then complete. Including those terminal identities as favorable predictive examples would hide the relevant limitation.

A separate four-state construction exposes a planning issue even when the restricted model is accepted. Let the two endpoint coefficient vectors be(1/4, -1/4, 0, 0) and(0, 0, 1/4, -1/4). The worst conditional width is1/2. Querying only state a or only state c leaves that worst width 1/2, whereas querying both reduces it to1/4. The marginal gain of c is therefore zero initially and 1/4 after a. This violates diminishing returns for the resulting robust width-reduction set function; a rule that stops on zero immediate gain can miss a useful pair.

The construction is algebraically compatible with zero-sum coefficient vectors and a remaining fraction 1/2: the implied active-frequency difference is(-1/2, 1/2, 1/2, -1/2). It is a deliberately authored counterexample, not a thirteenth natural history or an observed common failure. It does not refute conditional adaptive-submodularity results [21]; it shows that our objective lacks a property that such a guarantee would require us to establish. Exact visible-set enumeration is a transparent small comparator for this gap.

Horizon	Preterminal views	Outside segment	Query-time views outside	Terminal membership
5	5, including t0	4	3 / 4	Trivial at t5
8	8, including t0	5	4 / 7	Trivial at t8

Table 21: Exact retrospective coefficient-model membership, not statistical coverage or an independent misspecification rate. Current and terminal coefficients are distinguished; only evaluation reads the latter.

29.2 Preserved v19: A late-query control improves final width and delays information

The new replay has 56 policy/horizon/cap configurations, 420 decisions and 1, 260 safe bound rows over three fixed panels. The prior 24 terminal contracts reproduce exactly. Author-written separate verification reconstructs all prefix occupations directly from raw unit intervals, enumerates 483 candidate subsets by bit masks, checks 1, 235 convex-breakpoint evaluations and independently confirms all eight arrival-oracle contracts. These are computations on the same exposed authored inputs, not independent scientific repetitions.

At B=8 and q=3, late-prefix queries at checkpoints six, seven and eight, reaches final width 1/24 with zero arrival-oracle regret, and certifies the unaffected positive contrast at checkpoint eight. Restricted one-query and restricted-set rules reach width 1/12, regret 1/24 and no certificate within the horizon. Their query checkpoints are four, five and six. The timing tradeoff is retained: normalized checkpoint width area is75/64 for late-prefix versus 37/32 for restricted-set. Narrowing the final interval by waiting does not establish a better anytime auditor or a reduction in actual worker latency.

The quote count remains three in both B=8/q=3 comparisons. The terminal gain therefore concerns when a fixed query allowance is used, rather than extra quote spending. The area is a declared discrete left-step summary: sum the safe width at checkpoints zero through B-1 and divide by B. It is not continuous-time expected recovery, elapsed worker latency or a sampling confidence sequence. FIFO at that same contract also has area 37/32 and width 1/12. A final-width-only ranking omits the period during which an auditor has less information.

The opposite boundary is visible at B=8/q=1. FIFO queries at t=1 and late-prefix at t=8; both finish at width 1/8, while their checkpoint width areas are 75/64 and 19/16. Waiting gives no terminal improvement there and has the worse area. At B=5/q=4, current disagreement, persistence and both restricted minimax rules tie at width 1/15 and an unaffected positive certificate at t=5. Prefix_abs and late-prefix use only three effective queries and remain at width 2/15. These caps are upper bounds, not a forced equal number of useful actions.

In all eight final-width contracts, persistence_abs and the two restricted minimax rules tie. Their decision sequences and areas can differ: at B=8/q=3, restricted-set improves the checkpoint area by1/192 relative to restricted-one without changing final width 1/12. Thus additional finite planning has a narrow timing effect in this bank but establishes no better terminal width. Candidate-set work is reported separately from cache quotes:483 nonempty sets are evaluated along primary decisions, excluding repeated invariance checks, author verification and the clairvoyant oracle. No hardware-speed or model-cost saving is inferred.

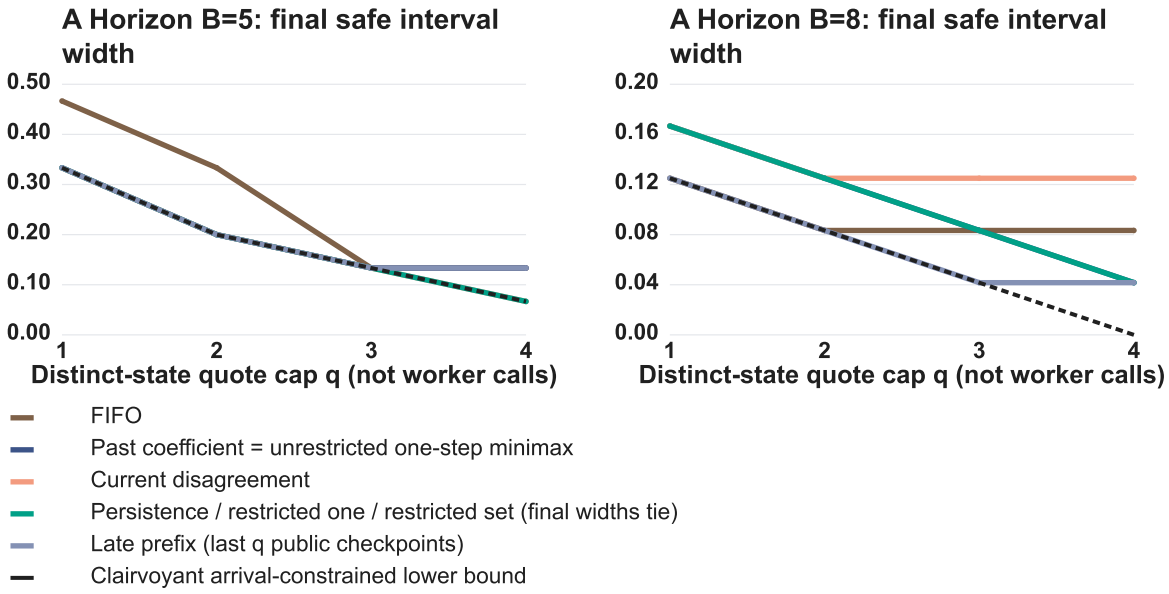
$$J = \frac{1}{B} \sum_{t=0}^{B-1} W_t \tag{57}$$

B / q	Policy	Final width	Quote checkpoints	Width area	Unaffected certificate
8 / 3	FIFO	1/12	1,2,4	37/32	Not reached
8 / 3	Restricted one	1/12	4,5,6	223/192	Not reached
8 / 3	Restricted set	1/12	4,5,6	37/32	Not reached
8 / 3	Late prefix	1/24	6,7,8	75/64	8
8 / 1	FIFO	1/8	1	75/64	Not reached
8 / 1	Late prefix	1/8	8	19/16	Not reached
5 / 4	Current difference	1/15	2,3,4,5	6/5	5
5 / 4	Restricted set	1/15	2,3,4,5	6/5	5

Table 22: Identical saved paths and quote opportunities. Certificates use unrestricted bounds; not-reached outcomes remain explicit. A checkpoint is a recovery-cost coordinate, not elapsed time.

SecondChance v17: conditional minimax is a comparator, not protection against every future

Same exposed authored logs / 12 templates / 56 dependent configurations. Actual new audit executions = 0.



B Does the actual final coefficient vector belong to the restricted continuation segment?

B=5	t0: out	t1: out	t2: out	t3: out	t4: in	t5: in		
B=8	t0: out	t1: out	t2: out	t3: out	t4: out	t5: in	t6: in	t7: in	t8: in

9 of 13 nonterminal views are outside (includes t0); these are dependent diagnostics, not a failure probability.

C At B=8 / q=3, later information narrows the final result but increases checkpoint uncertainty

Policy	Query checkpoints	Final width	Width area	Positive certificate
FIFO	1,2,4	1/12	37/32	not reached
Restricted-set	4,5,6	1/12	37/32	not reached
Late prefix	6,7,8	1/24	75/64	t8 / 3 quotes

Figure 9: New v17 robust-allocation comparison. Final unrestricted widths versus quote cap for horizons five and eight; the restricted model excludes the actual terminal vector in nine of thirteen nonterminal views. The B=8/q=3 table retains late-prefix final-width improvement with its higher checkpoint uncertainty area. Persistence and both restricted minimax policies tie on all eight terminal-width contracts, so their coincident curves are explicitly grouped. Counts are finite authored diagnostics, not model trials.

29.3 Preserved v19: Comparators, safety bounds and a prospective robustness test

The official revised adaptive-submodularity paper defines policies from partial observations and conditions greedy guarantees on structural assumptions [21]. We reopen its primary HTML policy and diminishing-returns sections; our robust counterexample is a direct local calculation, not a reproduction of that paper. We also reopen official COCO assessment guidance [16] to preserve fixed-budget quality versus target-attainment distinctions. Its optimization benchmark and reported performance are not executed or imported. The absent-target entries here remain null with query spending retained.

A clairvoyant arrival-constrained oracle sees the completed coefficient vector, obeys the same available state arrivals and one-query schedule, and lower-bounds the final residual width. It is not an online competitor with the same information. Unrestricted minimax equivalence applies to one current query against arbitrary future states, while restricted minimax applies to a declared continuation segment. Neither statement proves competitive online regret, calibrated confidence coverage or robustness of the actual complete-state assumption.

The current safety contract still assumes complete visible commits, full state identity, deterministic fixed-input panel utilities and no hidden mutation. Restricted-model misspecification is diagnosed without invalidating the separate unrestricted suffix bound inside that contract. In an actual stochastic or partially hidden executor even that underlying contract must be tested; an exposed arithmetic fixture cannot certify it. New actual audits, worker calls, natural histories and independently reserved prospective streams remain zero; prospective audit effect is null.

A next prospective test should freeze policy and continuation class before new commits arrive, log the exact selector view and available opportunities, and compare late/FIFO controls alongside any robust optimizer. It should report the final safe width, the checkpoint uncertainty area, target-attainment time, spent and failed quotes, cache hits, model/worker invocations and elapsed latency separately. It must audit full-state/input repeatability and reject deterministic reuse when hidden state or drift is observed. The present unfavorable model coverage and comparator results are admission tests for that design, not proof that robust auditing is useless in other settings.

29.4 Preserved v19: Unknown continuation scenarios change the audit objective

The continuation distinguishes the worst width conditional on a fixed scenario, the union of identification intervals when the scenario itself is unknown, and a pre-outcome worst union objective used for utility-blind audit selection. The last has four endpoint-pair cases; a two-state analytic construction separates zero conditioned width from union width1/2. This mathematical construction must not be substituted for observed incidence.

On all1,620 actual fixed-panel views in the saved development traces, actual union width equals worst conditioned width. The empirical premium is zero even though pre-outcome counterfactual objectives can differ. A supplementary envelope check enumerates1,104 visible subsets and11,159 known-utility vertices; unreachable historical audit subsets are explicitly separated from causally arrival-feasible choices.

Two new union-aware deterministic rules use the same visible prefix, arrival opportunities and quote caps as the seven preserved controls. All16 new-rule/cap paired contracts tie their corresponding restricted rules in final safe width. Four B8 areas improve by exactly1/192 at unchanged query checkpoints and quote counts; twelve areas tie. These repeated authored replay contrasts are not independent gains, actual elapsed audit savings or new unaffected-state certificates.

Quantity	Result	Interpretation
Actual panel union premium	0 /1620 views	No observed positive premium
Final paired widths	16 /16 ties	No final-width gain
Checkpoint area	4 decreases by1/192	Dependent replay only
Remaining areas	12 ties	Preserved neutral outcome

Table 23: The observed negative result and the counterfactual objective are kept separate.

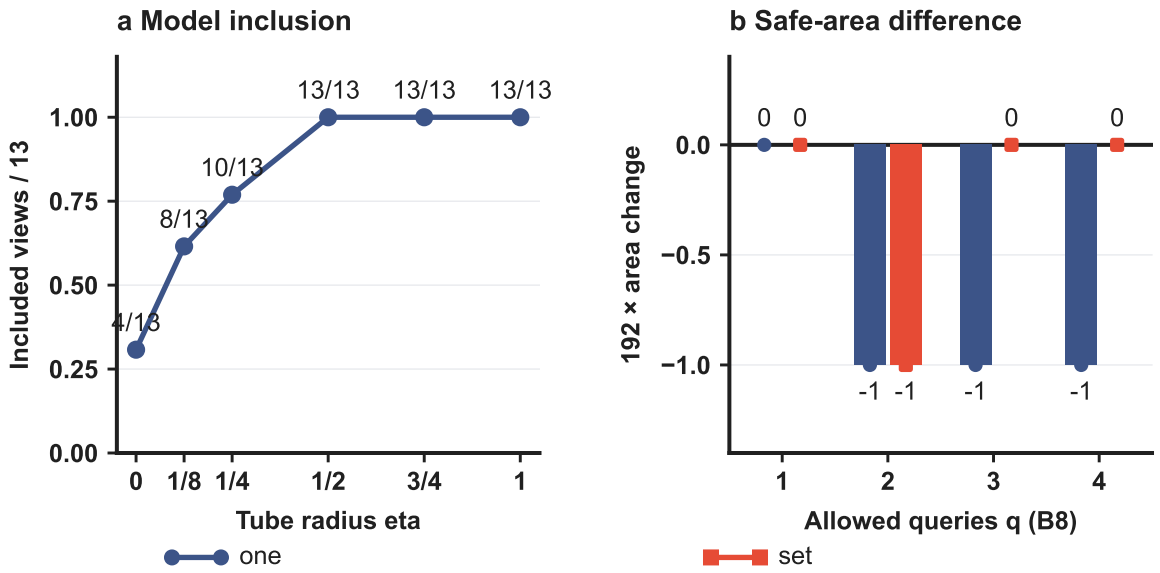


Figure 10: Left: retrospective inclusion among the same13 dependent nonterminal views as the declared tube radius changes. Right: B8 union-minus-restricted safe-area differences scaled by192; B5 differences are all zero. Inclusion is descriptive model fit, not confidence coverage, and the small area changes are cached replay outcomes rather than new real audits.

29.5 Preserved v19: Uniform ambiguity inflation is not a selector repair

Allow zero-sum coefficient perturbation with L1 norm at most epsilon=2 eta r, with freely valued fresh-state coordinates attaining0 and1. Under this declared unrestricted fresh-state tube, each relevant width and the pre-outcome objective increases by exactly epsilon. The positive/negative perturbation masses are epsilon/2. If the same eta and remaining mass are used for every candidate, inflation adds the same constant and cannot change the candidate ranking. A tighter physically attainable continuation set would require a different analysis.

At eta0,1/8,1/4,1/2,3/4,1, the final coefficient is included in4,8,10,13,13,13 of the same13 dependent nonterminal views. Selecting eta from these final outcomes would not create prospective calibration. The safe unrestricted bound remains the only certificate in the replay; restricted widths are conditional diagnostic objectives.

At B8/q3, late retains final safe width1/24 and area75/64; FIFO, restricted-set and both union rules have width1/12 and area37/32. A two-quote current-disagreement control remains nondominated on the declared width/area/quote axes. The frontier ranges over nine declared finite rules and cannot certify a globally optimal causal audit policy or a real latency frontier.

30 Preserved v19: Discussion and Limitations

The practical lesson is a decision rule: do not promote a recovered score, candidate or skill until the integrated state has been observed under the corrected contract and preserved capabilities have been checked. This is a measurement requirement supported by constructed counterexamples. It is not evidence that the proposed lineage-first procedure should replace the neutral control.

30.1 Preserved v19: Identification assumptions and threats to validity

Construction is the dominant limitation. One author defines arithmetic inputs, candidate programs, the evaluator defect, lineage and harmful lesson routing. Eight histories contain a useful rejected candidate and four do not. Those proportions, the aligned/misleading balance and the recovery counts are experimental choices and consequences, not deployment incidence estimates. The same few inputs recur throughout; researcher-visible audit answers prevent any independent final-test interpretation.

Complete-state reconstruction is conditional. Files, retrievers, caches, tool responses, external services and latent or stochastic model state could invalidate the claim that utility stays fixed between logged commits. A same-state disagreement, an unknown state mutation, a missing positive-duration audit or a violation of the bound must suspend a point estimate. For stochastic agents, state-level repeated

sampling and a declared expectation/uncertainty model are needed; one observed answer does not identify capability.

The later-input panel runs the same final state on a different authored input. No further training, candidate evolution or calendar-time drift occurs. The historical field strict_recurrence is therefore a maintenance-failure statistic across these panels, not long-term evolutionary recurrence. Similarly, exact local hashes support integrity checks, not external authorship signatures, scientific correctness or conference acceptance.

30.2 Preserved v19: What fails without complete logs or deterministic equality

The positive identification results retain complete commit times even when utility audits are withheld. If that condition is removed and arbitrary hidden continuous-time state changes are allowed between finite checkpoint observations, those observations do not bound the area away from zero or one. In every gap we construct a path matching the left observation for a gap/1000 neighborhood, a low or high utility in the interior, and the right observation near the next checkpoint. The two paths agree at every retained audit but have normalized areas at most 0.002 and at least 0.998. The construction is checked for all 504 trajectory-panel objects. The sharp closed region is therefore the full unit interval as the neighborhoods shrink; exact endpoint attainment under right-continuity is not asserted.

This counterexample deliberately removes an integer-only commit lattice and permits otherwise unrestricted hidden-state utility dynamics. If a documented lattice, bounded number of hidden events, minimum dwell time, known transition graph or restricted worker semantics remains available, tighter bounds can follow and must be optimized over that smaller class. We do not claim that every kind of partial logging is vacuous. Nor do these compatible mathematical paths constitute newly executed recovery trajectories or additional natural histories.

Deterministic state equality is a separate assumption. If an apparently identical hash omits environment state, or stochastic outputs are treated as fixed, neutral and selective visits can have different utilities. Equal occupation coefficients then no longer cancel. Two visit-specific utilities zero and one yield a contrast of minus one, while reversing them yields plus one, despite a zero contrast under the shared-variable model. Repeated state audits, environment snapshots and an expectation/uncertainty contract are needed for stochastic agents. A checksum cannot supply them.

30.3 Preserved v19: A staged real-model confirmation study

The executed negative findings narrow the proposal. Lineage-first ordering should be a conditional hypothesis about trustworthy decision support, with neutral fallback and its cost included. A useful-rejected-candidate gate before integration may address development memorization, but its extra queries, exposure and selection opportunity must be charged. Reusing the final audit labels for that gate would change the evaluation rather than demonstrate improvement. Current-only model retuning should be added as a separately budgeted generation procedure, not described as a native comparison already completed here.

A model feasibility study needs an actually generated versioned candidate archive, real old-grader accept/reject decisions and any lesson the model really wrote. Before correction, freeze the common history, active artifact, preserved later changes and model/runtime/settings. The correction needs independent approval under an unchanged contract. Assign protected task roles and stop rules before generating outcomes; keep no-useful-reject, no-written-lesson, disputed-correction, missing-archive, failed-integration and incomplete-audit histories visible in the flow.

Matched-policy forks must have common data rights and envelopes for selector calls, candidate invocations, input/output tokens, grader calls, integration, lesson editing and audits. A complete model state may include filesystem, tool responses, retrieval indices and environment snapshots beyond a prompt hash. For stochastic outputs, repeated state audits estimate a panel expectation with nested sampling uncertainty; one deterministic fixture output supplies neither a population confidence interval nor a stochastic capability curve. New correction families and evolution prefixes must be clustered rather than treating repeated responses as independent histories.

Only after those conditions are met can the hypotheses concern real recovery or continued-evolution recurrence. A native SkillOpt, Reflexion or Zep comparison additionally needs the original algorithm's declared objective, environment, reset and resource contracts. Until then, our findings concern endpoint design, a fully inspectable constructed failure mechanism and measurement resolution. Model critic feedback, named methodological lenses and successful checksum verification do not close those scientific gates.

30.4 Preserved v19: Benchmark choice and falsifiable next hypotheses

A feasible first extension is an access-controlled program-edit history over a fixed subset of real issue-resolution tasks. SWE-bench can motivate the environment [18], while EvalPlus motivates additional behavioral tests [17]; neither alone supplies correction labels. Before model use, construct an explicit inclusion flow: unchanged requirement, authenticated old rejection, independent correction approval, available candidate archive, observable current state, and approved audit rights. Preserve failed, disputed and absent-useful-candidate histories in the flow. Repository licenses, runtime feasibility and evaluation exposure must be checked before acquiring or executing a selected dataset.

The primary superiority hypothesis should be conditional on reliable lineage and tested against neutral early-stop, neutral joint repair and a separately budgeted current-only regeneration procedure. A useful-candidate gate must use permitted development data and pay for its extra queries; reusing final audit labels would change the task. The primary recovery endpoint should be fresh integrated-state utility on independently held histories, with preserved-capability loss and total model/tool resources as co-primary costs. The stopping rule, smallest relevant effect and interval procedure should be set before model outcomes, rather than obtained by treating 168 nested paths as independent observations.

Long-term recurrence requires another declared evolution prefix after repair, with new decisions, lessons and resource/exposure records. It is a separate future experiment. The current design will be rejected or revised if neutral procedures retain their advantage on the declared target, if lineage fails to localize useful rejections, if the state snapshot cannot support reproducible audits, or if additional intervention cost outweighs any independently measured behavioral gain.

31 Preserved v19: Conclusion

Evaluator correction, candidate success and current-state recovery are different endpoints. Second-Chance makes the distinction executable and exposes a strong neutral control that ties recovery and can avoid unnecessary cost and collateral loss. Its frozen-path follow-up shows that coarse utility sampling can miss both recovery and degradation; under complete deterministic state logs, freshly observed commits identify a stepwise area with an explicit geometry bound. The supported contribution is an auditable measurement and comparison contract with retained negative results. Natural LLM recovery, native-method superiority and long-term recurrence remain unmeasured.

Controlled censoring adds a precise distinction: a policy contrast may remain exactly or one-sided identified even when separate recovery areas are not. Signed state-occupation bounds and exhaustive audit-set certificates quantify that robustness in this exposed fixture. The same analysis exposes its failure boundaries: unknown commit timing and invalid deterministic-state equality can destroy identification. These are measurement findings and future audit requirements, not a demonstrated real-agent recovery advantage.

V15 supplies a concrete utility-blind audit order for the known-path setting and verifies its exact width objective. Its five/four-query exact certificates and four/three-query positive-sign stopping costs must remain distinct from hindsight minima, true online scheduling and actual audit latency. Natural correction histories and complete deterministic state observations remain the empirical admission gates.

V16 executes decisions without future commits and bounds the unknown suffix explicitly. Causal replay reveals different winners at different horizons and query caps, while a same-prefix/opposite-future construction prevents universal early sign certification. The advance is an executable information contract and an exact finite comparison; prospective acquisition cost, natural correction-policy effects and broader scheduler guarantees remain open.

V17 adds a conditional minimax objective, an exact misspecification check and a delayed-query control. Stronger-looking optimization does not automatically beat simple scheduling: late-prefix attains a narrower B=8/q=3 final interval than both restricted minimax rules, but waits longer with higher checkpoint width area. Conditional continuation robustness, unknown-scenario identification and actual prospective audit efficiency remain distinct claims. The latter still requires new independent instrumentation and natural histories.

V19 separates scenario-conditioned, realized scenario-union and pre-outcome union objectives. The observed premium is zero in all 1,620 panel comparisons. Across the sixteen paired new-rule contracts, final safe widths all tie; only four checkpoint areas improve, each by 1/192. A zero-sum L1 continuation tube adds an exact common width term under stated free-coordinate conditions and therefore cannot alter candidate ranking. Increasing empirical inclusion is a descriptive sensitivity result, not a coverage guarantee. The next test requires prospectively reserved paths and measured audit costs.

Acknowledgements
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A Reproducibility and failed attempts

The v11 source is projects/research_progress_v11_20261018/second_chance/lab.py and its original execution is runs/research_progress_v11_20261018_01/second_chance. The historical v12 state re-audit runner is projects/research_manuscripts_v12_20261018/second_chance/commit_audit.py. Its –out command requires a fresh directory, verifies prior sealed inputs and invokes the existing authored local worker. The new v14 censoring runner is projects/research_advancement_v14_20261019/second_chance/analyze.py; it reads preserved observations without invoking that worker. Neither analysis requires a model endpoint, installed dependency or an active-skill change, and neither overwrites an original run.

The complete follow-up is runs/research_manuscripts_v12_20261018_01/second_chance/commit_audit_02. Its plan.json fixes inputs and assumptions; traces/ records reconstructed commits and state-bound fresh observations; invocation_receipts.json binds subprocess requests/outputs; results.json retains every policy/budget denominator; receipt.json hashes every output and consumed input. commit_audit_01 preserves the failed adapter attempt and its exact old analysis bytes. Unsigned hashes supply local integrity rather than publisher identity, provenance signatures or scientific certification.

The historical v12 read-only verifier, projects/research_manuscripts_v12_20261018/second_chance/verify.py, recomputes input/output hashes, state reconstruction, deterministic output equivalence, audit costs, coarse and commit-aware areas, geometric bounds and aggregates without invoking subprocesses. Seven authored metric self-checks cover a tight one-fifth discrepancy, grid-aligned commits, positive-width missingness, terminal-only missingness, decreasing utility and unchanged-state events. Four completed-run checks require all 168 reconstructions, concordance, geometry bounds and all twelve initiated templates. Additional tamper/path rejection validation is recorded separately; no fabricated independent experiment is created by a test assertion. The separate v14 verifier is projects/research_advancement_v14_20261019/second_chance/verify.py: it checks the new frozen snapshots, independently aggregates raw-segment means and reconciles controlled-censoring bounds without new worker execution.

B Proof details and missing observations

Fix an audit input and suppose every state-changing event is logged; the executor is a pure deterministic function of the complete recorded state; and every state occupying positive cost width has a fresh valid audit. Partition [0,B) at the state commits. State and input remain constant on each half-open interval, so utility is constant and its integral equals utility times interval width. Summing yields A_C. Endpoint-only states require a separate terminal observation even though their contribution to area is zero.

For a coarse interval [g_j,g_(j+1)), a grid-aligned commit is included in the state at its left endpoint. If the first strictly interior state-changing commit is t_j, actual and left-sampled utilities are equal on [g_j,t_j). On the remaining interval their difference has absolute value at most one. The absolute integral error is therefore bounded by its length; the triangle inequality over cells and division by B gives E. Extra commits can cancel error but cannot violate this bound. With utilities in [L,H], multiply E by H-L. A single persistent jump at t_j attains the bound.

For missing positive-width observations, let W_unknown be their total width and sum only known contributions as A_observed. With utilities in [0,1], area lies in [A_observed, A_observed + W_unknown/B]; retain null for an unidentified point estimate. Missing endpoint-only audits do not create area width and do not authorize filling in a terminal score. Unknown commits or incompletely captured environments invalidate the partition itself, rather than merely increasing a conventional standard error.

C Source review and reporting boundaries

SOURCES.json records nineteen primary-source reviews performed on 2026-10-18, their reviewed sections, short retained excerpts, excerpt hashes and claim links. A hash binds the retained excerpt and local review record, not a publisher webpage, DOI identity or an unsigned scientific truth claim. Most methodological comparisons use official abstracts; selected full-text methods were additionally inspected for SkillOpt, Reflexion, Self-Refine, CRITIC and Zep. The Huang arXiv HTML retrieval failed, so its scope is restricted to the official ICLR abstract. No arbitrary remote PDF was downloaded.

The v12 manuscript cited a mutable SkillOpt-Sleep companion preview. It is omitted from this revision because it was not necessary to identify a new scientific comparison beyond the version-

pinned SkillOpt method. Reflexion conference metadata and arXiv v4 have different author lists; the bibliography uses the official proceedings edition and identifies the separate method-review edition explicitly in SOURCES.json. No published performance number is used as a locally reproduced result. No venue-specific submission format or acceptance claim is implied by this neutral conference-style preprint.

D Censoring artifacts, sharpness proof and disclosure

The new project is projects/research_advancement_v14_20261019/second_chance; the complete output is runs/research_advancement_v14_20261019_01/second_chance/censoring_01. PLAN.json, analyze.py and test_bounds.py were copied before omission outcomes were computed. source_receipt.json identifies 177 consumed source files and their byte snapshots, including 168 prior traces; receipt.json seals the new outputs. The earlier v11/v12 runs and v13 originals are unchanged. audit_subset_bounds.jsonl contains 3072 exact interval records; contrast_views.json and policy_area_views.json contain six predeclared disclosure views; certificates.json records retrospective minima; ambiguity_witnesses.json preserves endpoint completions with opposite signs; hidden_commit_counterexamples.json records 504 compatible low/high constructions; state_catalogue.json discloses the full nine-state oracle used to assess controlled omission. Fraction strings preserve exact endpoints, while float fields support plotting.

For sharpness, keep every revealed score fixed. Set each unknown state utility to zero for a positive coefficient and one for a negative coefficient to attain the lower endpoint, and reverse those assignments for the upper endpoint. Each state is assigned once, preserving all cross-policy equality constraints. The assignment is valid for the marginal binary-utility information model. If code semantics or joint-panel restrictions add constraints, this box optimization may become conservative for that stronger model.

Independent raw-segment verification reaggregates mean areas from the frozen traces rather than calling the analysis occupation function. Boundary tests include missing utilities, repeated-state and terminal-zero-width commits, fractional commit times, malformed geometry, signed extrema, equality cancellation and the distinction between weak and strict dominance. The analysis uses no new worker execution, network request or model API; primary-source browsing for the added reference is recorded separately.

The partial-identification citation was reviewed only at the official publisher metadata and abstract. It motivates separating compatible-value bounds from sampling confidence; neither its technical confidence procedure nor its asymptotic guarantee is implemented. All nineteen v13 reference roles and their bounded inspection records are inherited with explicit hashes, not represented as newly re-read full texts.

E V15 known-path scheduler replay inventory

The current manuscript, guide, PLAN.json, analyze.py and verify.py are in projects/ research_advancement_v15_20261019/second_chance. The immutable consumed plan/code and168 copied traces are in runs/research_advancement_v15_20261019_01/second_chance/analysis_01. input_lock.json precedes computation; query_paths.json contains180 exact-bound rows; results.json contains schedule orders,18 stopping summaries and retrospective certificates; receipt.json binds their hashes.

Execute with Python standard library only into a fresh directory: python projects/research_advancement_v15_20261019/second_chance/analyze.py –out runs/ research_advancement_v15_20261019_01/second_chance/NEW_RUN. Run verify.py at the same project path for the recorded run. No new worker/audit invocation or natural-history generation is hidden behind simulated query counts.

F V16 prefix-only replay inventory

PLAN.json and analyze.py are frozen with verify.py and all 168 original trace bytes before result generation. The new analysis directory contains visible_views.json, decisions.json, bounds.json, oracle.json, unknown_future_witness.json, results.json, input_lock.json and receipt.json. A new verification_01 directory records the separate-formula checks. The verifier is by the same author and does not constitute independent peer review.

The exact target retains twelve equal-weight authored templates in each arm. We snapshot all policy paths for provenance but do not treat 168 files, 48 contrast records, 24 configurations, 180 decisions or 540 bounds as independent samples. Cache quote values are old saved observations. Actual new state executions, worker invocations, natural histories, model fits and model API calls are zero.

G V17 conditional-allocation execution and verification inventory

The new analysis root is runs/research_advancement_v 17_20261019_01/second_chance/analysis_ 01. Input locking copies all 168 v16 trace snapshots, PLAN.json, analyze.py, verify.py and two supporting artifacts before numerical analysis. The frozen v16 helper supplies the unchanged sanitized view, baseline rules, interval formula and arrival-constrained oracle; its bytes are copied to runtime/ v16_prefix.py and hashed as an input. The old v16 result is a separate reproduction target. No imported run function is executed.

The input_lock schema adds a role field to distinguish 168 authored_trace records from two frozen_support records. It otherwise keeps path, snapshot, bytes, sha 256 and plan/code/ verifier bindings. New outputs are visible_views.json, decisions.json, bounds.json, oracle.json, model_membership.json, complementarity_witness.json, results.json and receipt.json. verification_ 01/verification.json records the same-author distinct computation. Root-owned nonauthor content review, typesetting and publication are separate.

Every primary decision record retains quote IDs, query count, restricted conditional objective and planned visible subset. Model membership uses full coordinate unions rather than checking only currently observed identities. A miss is retained even if an observed utility contrast happens to agree. Three old fixed panels reuse the same coefficient geometry and do not create additional independent paths. The figure receipt binds 56 terminal configurations and 15 membership views to the exact result hash.

H Source inspection boundaries

[1] Author arXiv abstract and HTML Sections 3.1/3.5; bounded edits, validation gate and rejected-edit feedback only. No native execution or superiority comparison.

[2] Official conference abstract; selected method Section 3 of arXiv:2303.11366v4 reviewed separately. Different author lists are not merged.

[3] Official abstract and arXiv:2303.17651v2 Section 2; same-model generation/feedback/refinement loop. Published effects are not imported.

[4] Official abstract and arXiv:2305.11738v4 Section 3/Algorithm 1; tool-derived feedback and output revision, not retrospective archive repair.

[5] Official abstract only: intrinsic reasoning self-correction without external feedback. HTML v3 retrieval failed; no full-text or universal impossibility claim.

[6] Official abstract only: SCoRe online multi-turn RL and self-generated correction traces. No assertion of failure under later methods.

[7] Author arXiv abstract: generate candidates and select using a verifier. No GSM8K execution or external performance transfer.

[8] Author abstract and HTML Section 2.2.3: temporal validity/edge invalidation. World changes are distinguished from fixed-requirement correction.

[9] Official abstract: proxy versus gold-reward optimization. No numerical scaling law is claimed for our deterministic fixture.

[10] Official abstract: overfitting a finite-sample selection criterion and biased performance evaluation. Motivates protected audit roles; no theorem transfer.

[11] Author abstract: adaptive reuse can overfit a holdout. No privacy mechanism or guarantee is implemented here.

[12] Author abstract and official AAAI search result; concerns nondeterminism, variability and experimental reporting. No RL experiment here.

[13] Author abstract: finite-run uncertainty and aggregate evaluation. We do not attach population intervals to designed templates.

[14] Publisher abstract/bibliography only: distribution functions of a performance metric. Not claimed as a recovery-area theorem.

[15] Author abstract: problem instances, target values and objective-call runtime. Resource-accounting motivation only; no optimizer benchmark.

[16] Official COCO HTML Introduction, Quality Indicators, Fixed-Budget versus Fixed-Target, Missing Values and Runtime Computation reviewed in v16. Target/nonattainment accounting only; no COCO implementation, objective-call speedup, restart or population comparison. V17 reopens official assessment HTML for fixed-budget versus target-attainment reporting; no COCO functions or timing comparison run.

[17] Official abstract: test augmentation and previously undetected errors. Does not establish natural evaluator-correction histories.
[18] Official abstract: repository issue resolution through code edits. Future environment candidate only; no acquisition, execution or correction-history labels.
[19] Author abstract: interleaved reasoning and environment actions. Motivates recording tool/ environment state; no direct recovery comparison.
[20] Official publisher metadata and abstract only: identification-region bounds and confidence coverage are distinct. No full-text theorem inspection, confidence algorithm implementation or sampling guarantee is claimed.
[21] Author HTML Sections 2-4 and scoped guarantee/revision notice reviewed. Partial-observation policy comparison only; no transferred approximation guarantee, Bayesian implementation or reproduced sensor experiments. V17 reopens primary HTML policy and diminishing-returns definitions; the local non-submodular robust objective is a separate counterexample and inherits no guarantee.